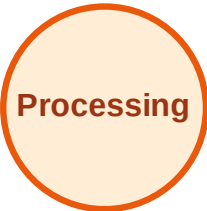


BFS Traversal

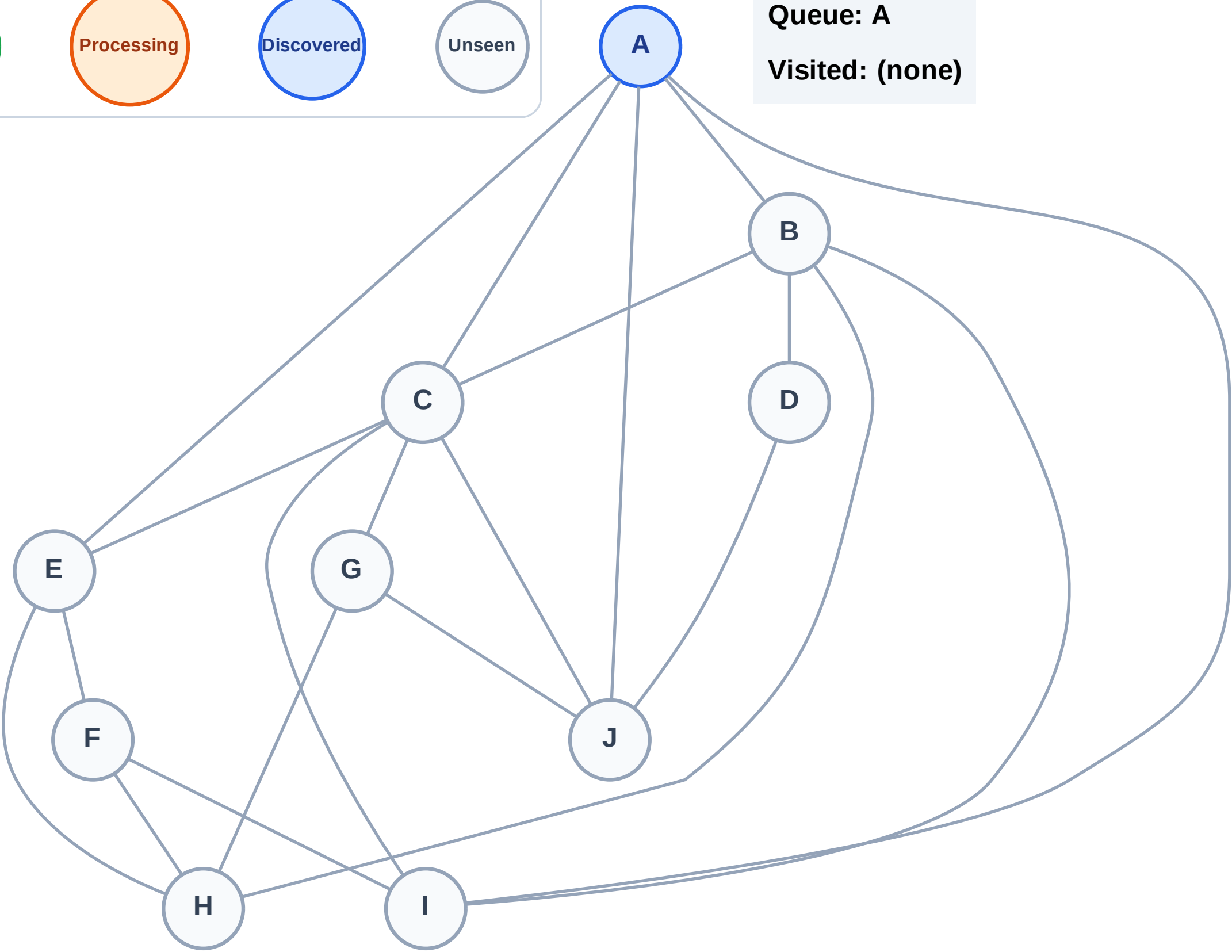
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

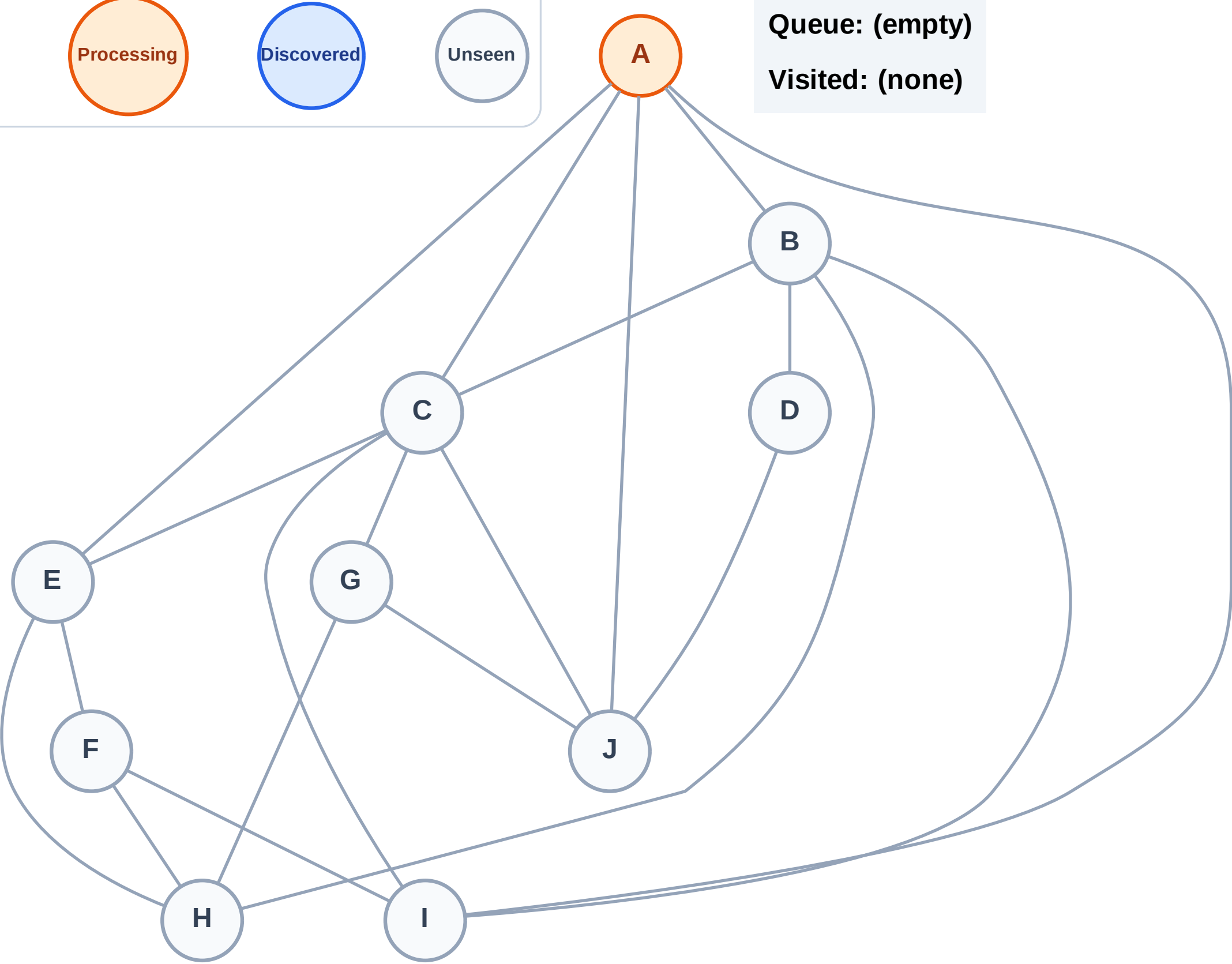
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, I, J from A

Legend

Complete

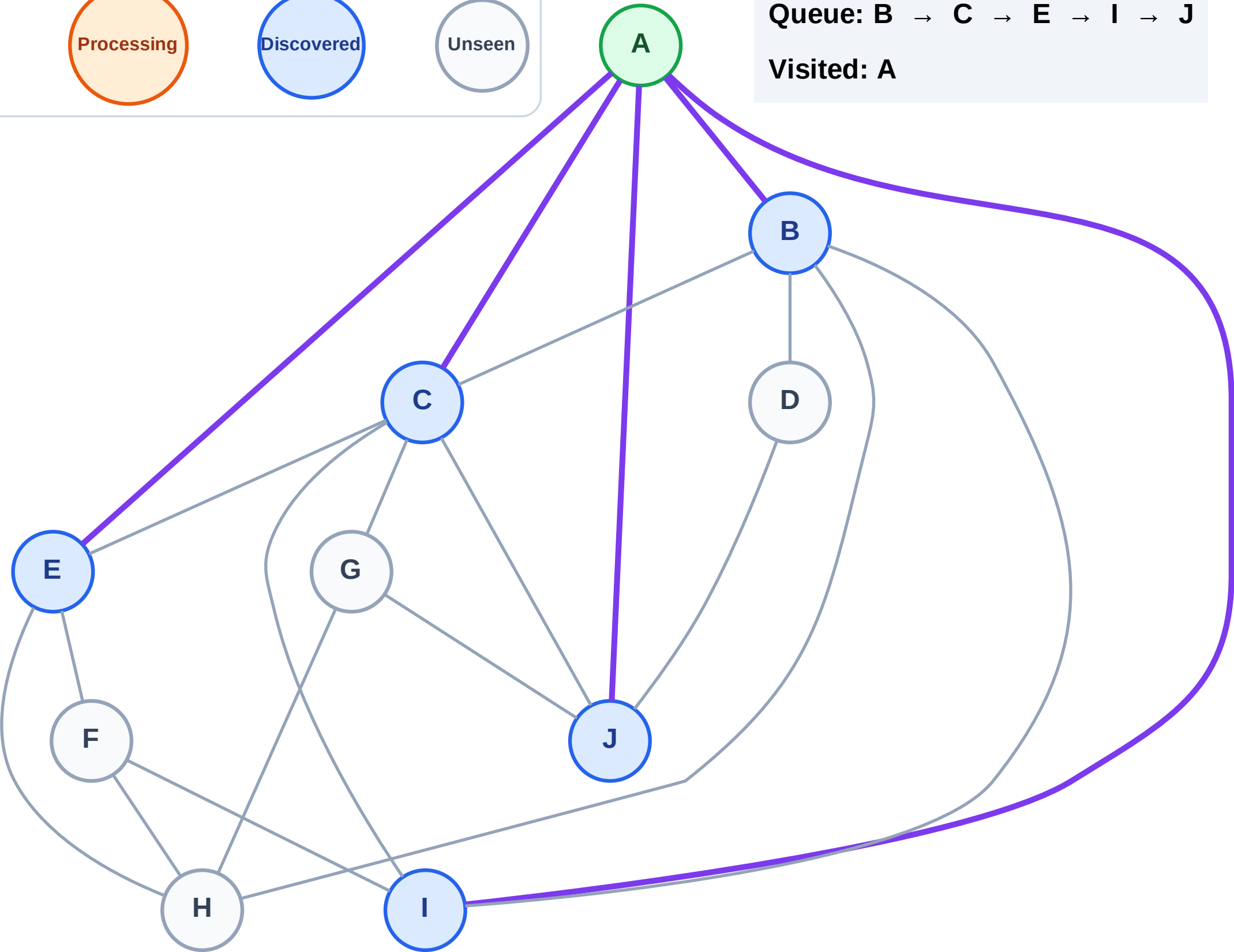
Processing

Discovered

Unseen

Queue: B → C → E → I → J

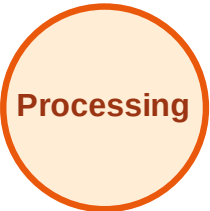
Visited: A



BFS Traversal

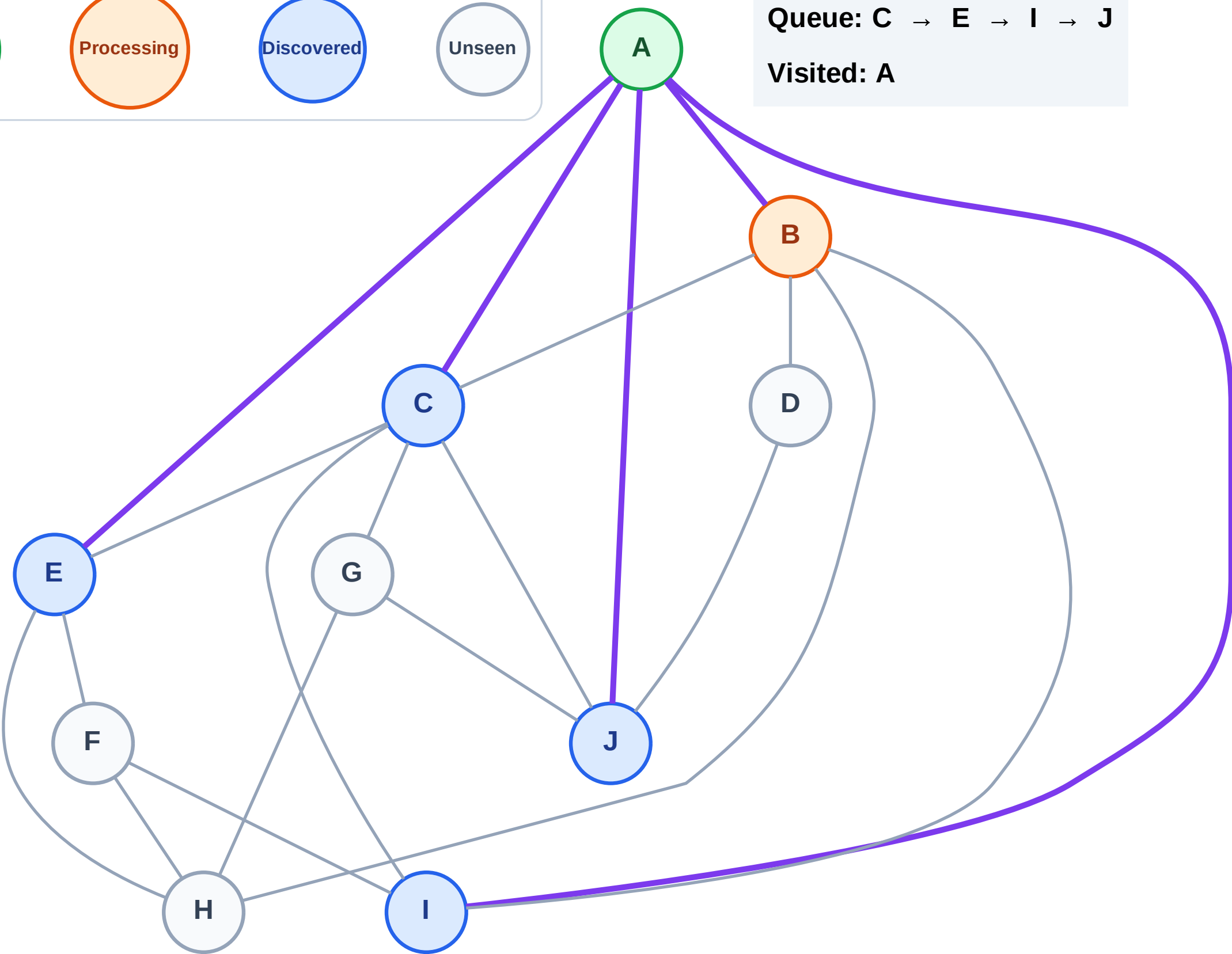
Step 4: Process node B

Legend



Queue: C → E → I → J

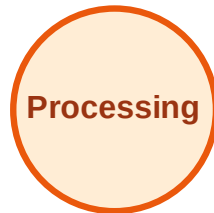
Visited: A



BFS Traversal

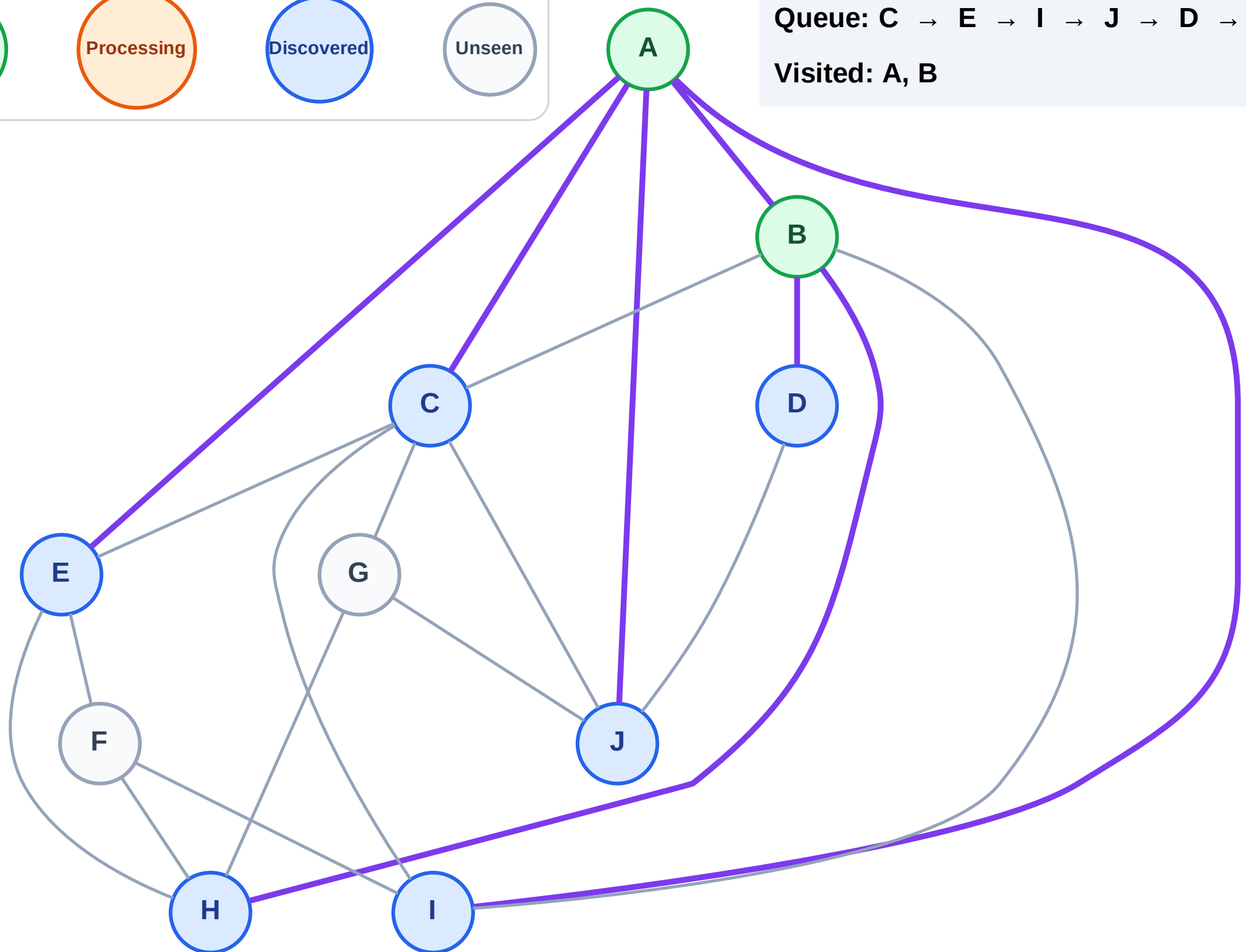
Step 5: Discover D, H from B

Legend



Queue: C → E → I → J → D → H

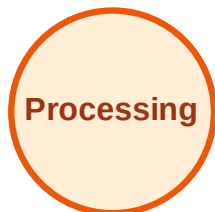
Visited: A, B



BFS Traversal

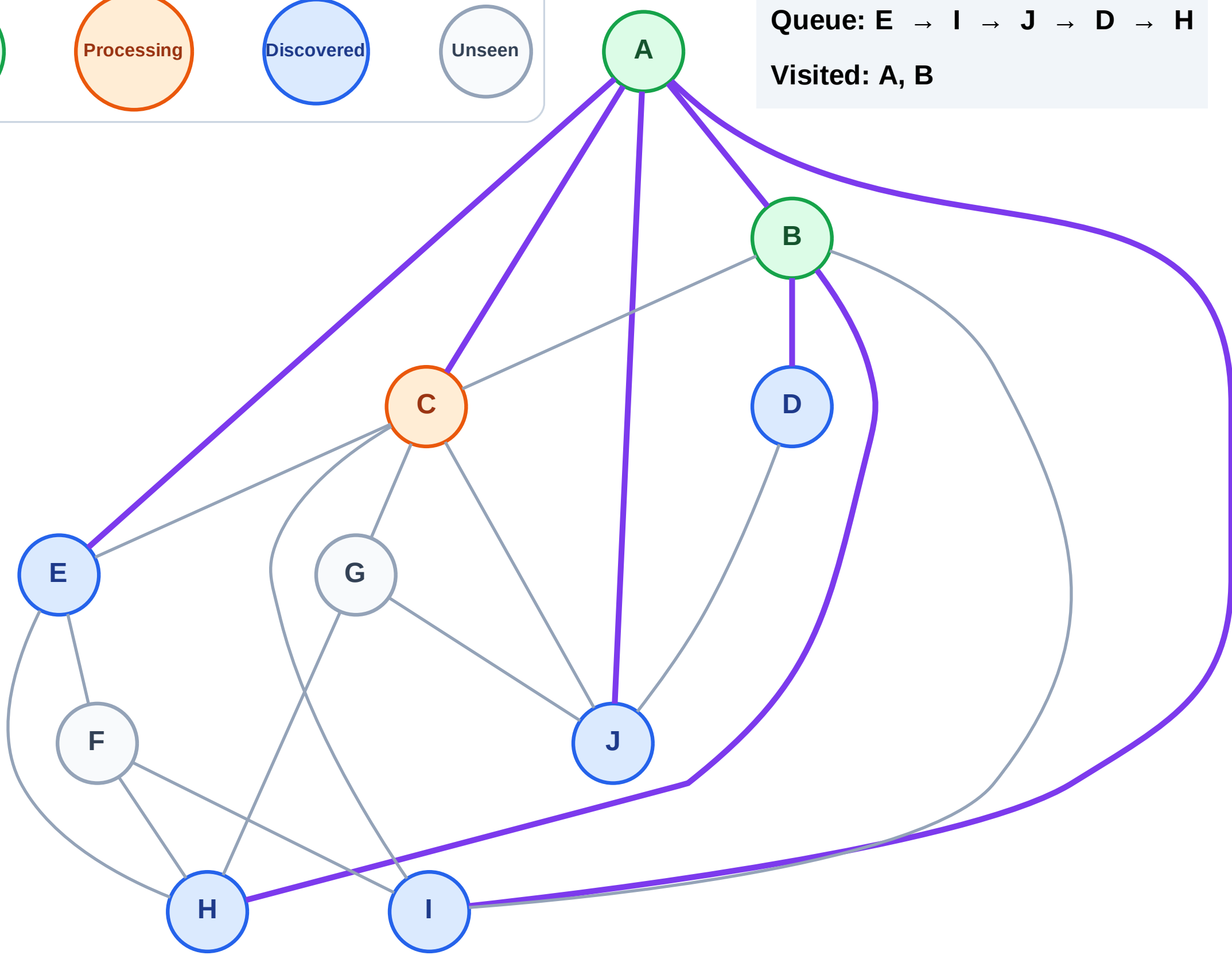
Step 6: Process node C

Legend



Queue: E → I → J → D → H

Visited: A, B



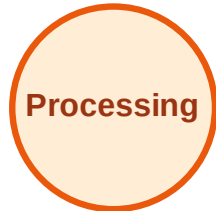
BFS Traversal

Step 7: Discover G from C

Legend



Complete



Processing



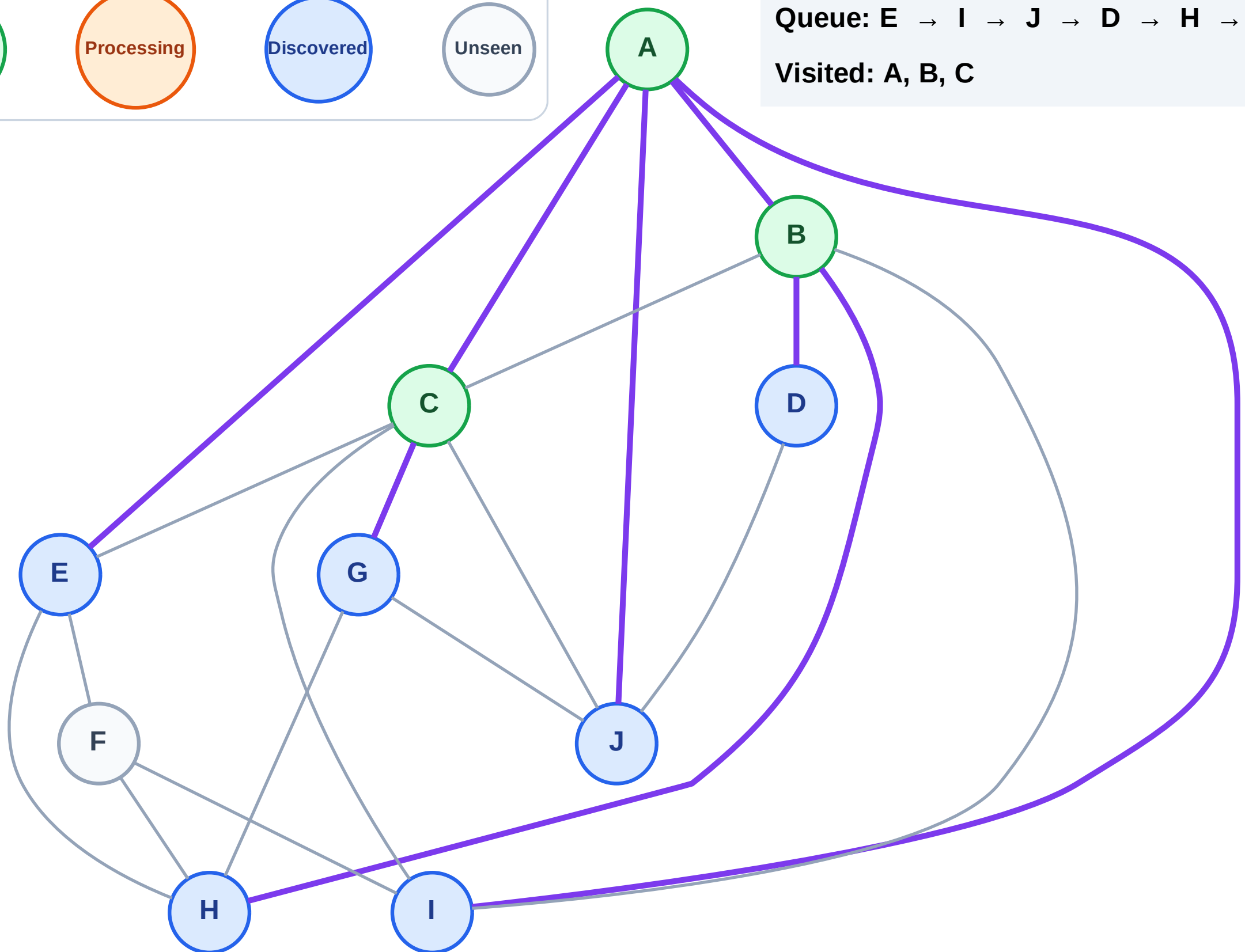
Discovered



Unseen

Queue: E → I → J → D → H → G

Visited: A, B, C



BFS Traversal

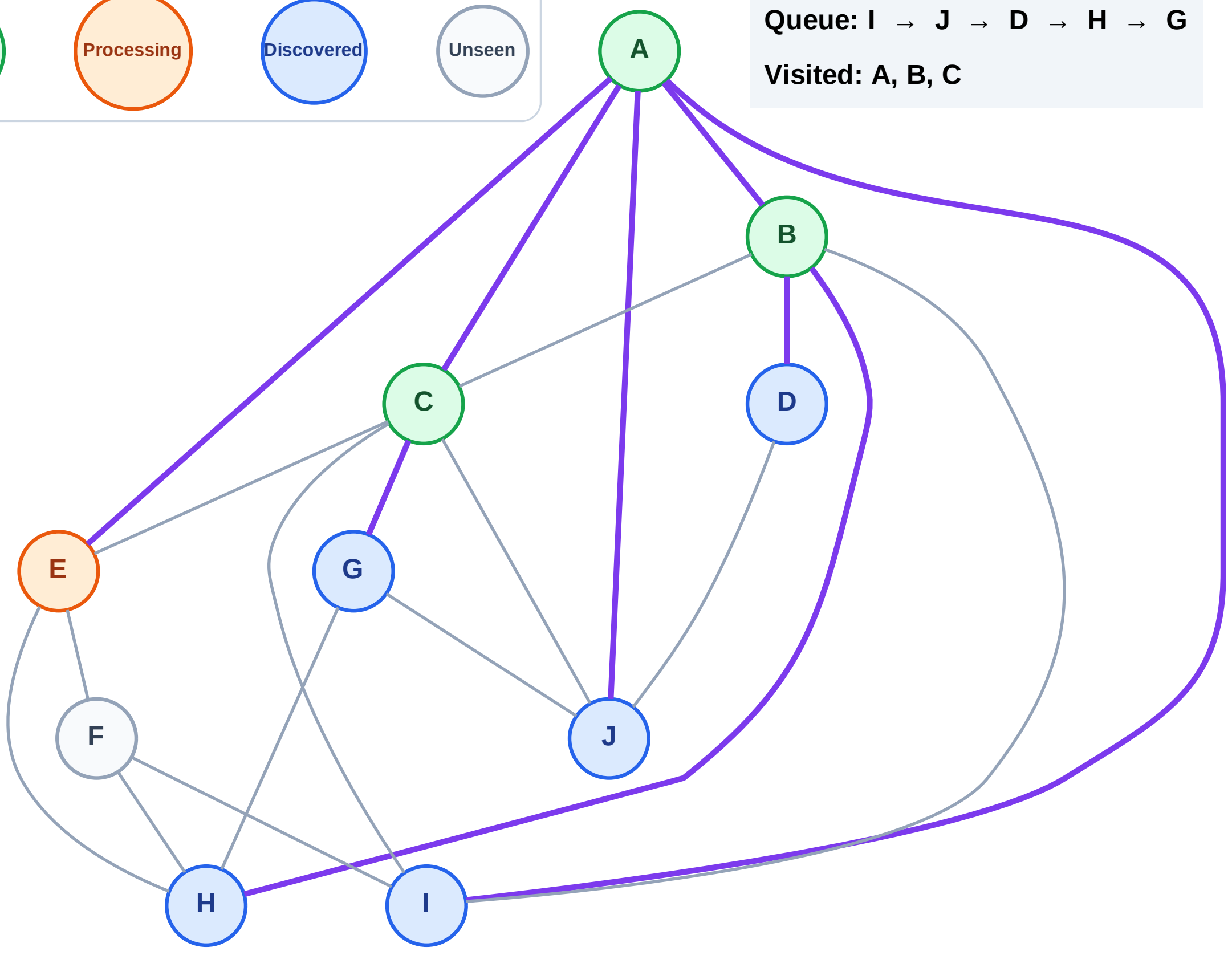
Step 8: Process node E

Legend



Queue: I → J → D → H → G

Visited: A, B, C



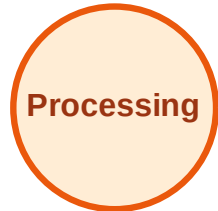
BFS Traversal

Step 9: Discover F from E

Legend



Complete



Processing



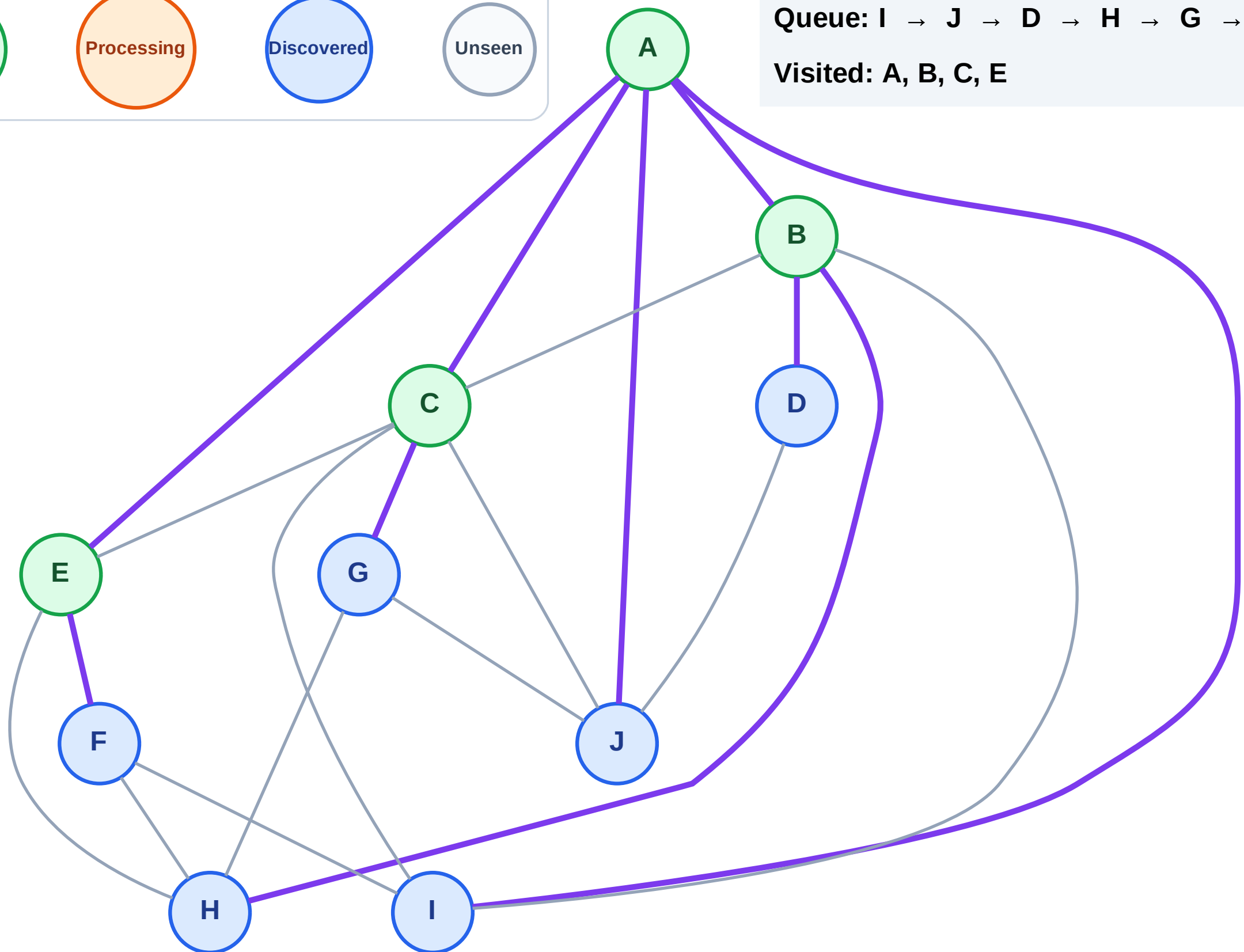
Discovered



Unseen

Queue: I → J → D → H → G → F

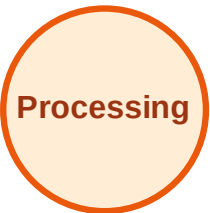
Visited: A, B, C, E



BFS Traversal

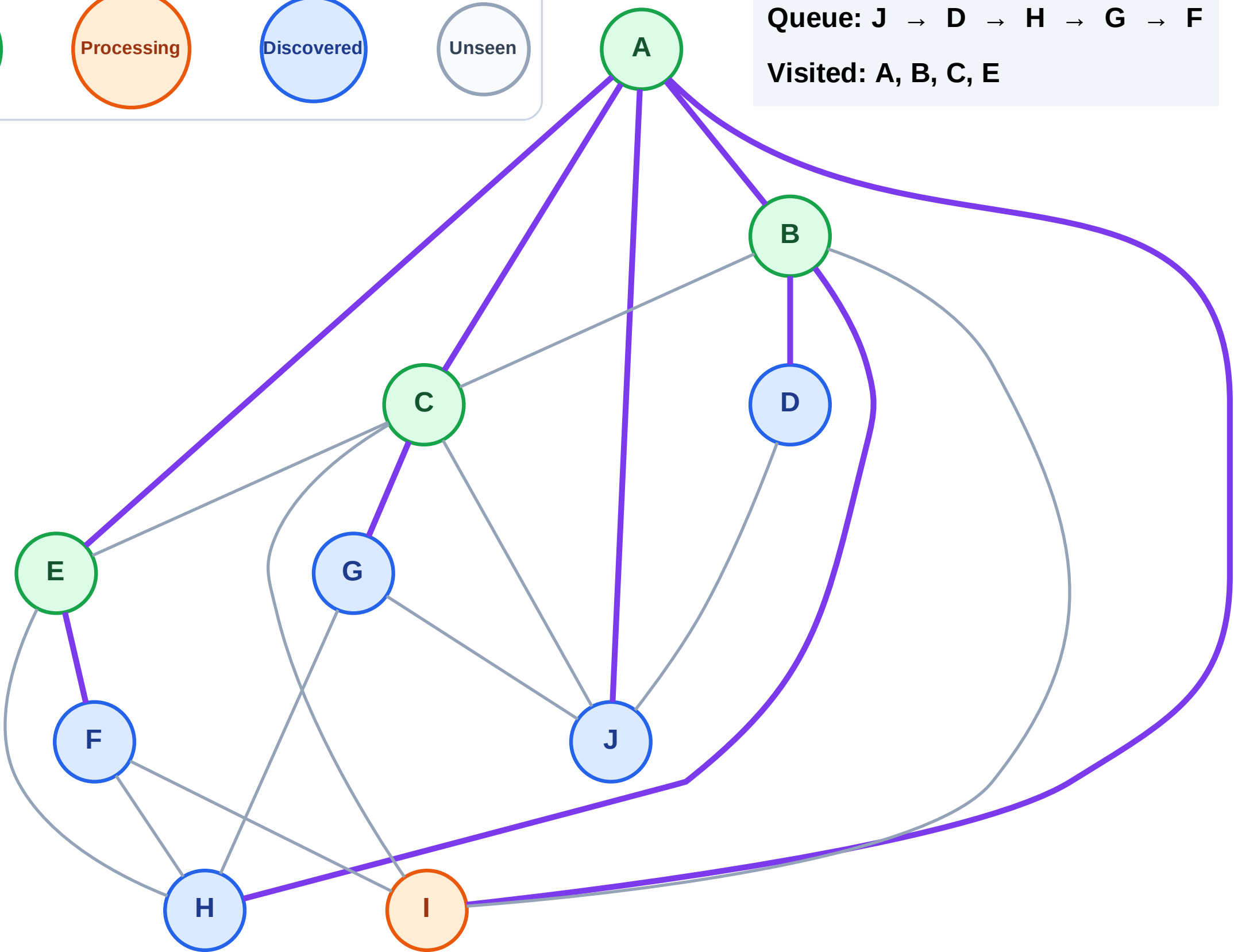
Step 10: Process node I

Legend



Queue: J → D → H → G → F

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from I

Legend

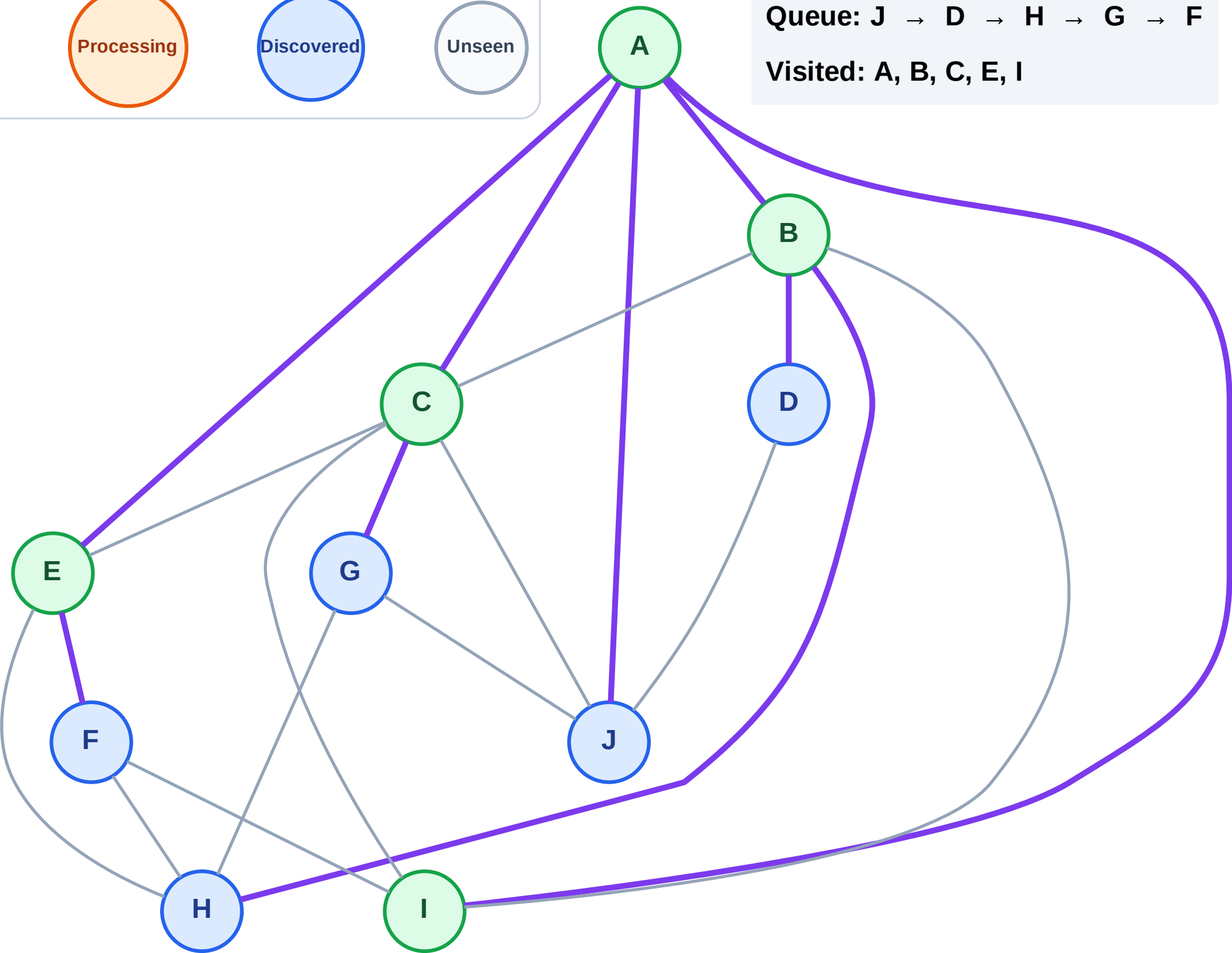
Complete

Processing

Discovered

Unseen

Queue: J → D → H → G → F
Visited: A, B, C, E, I



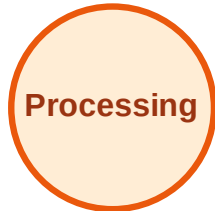
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing



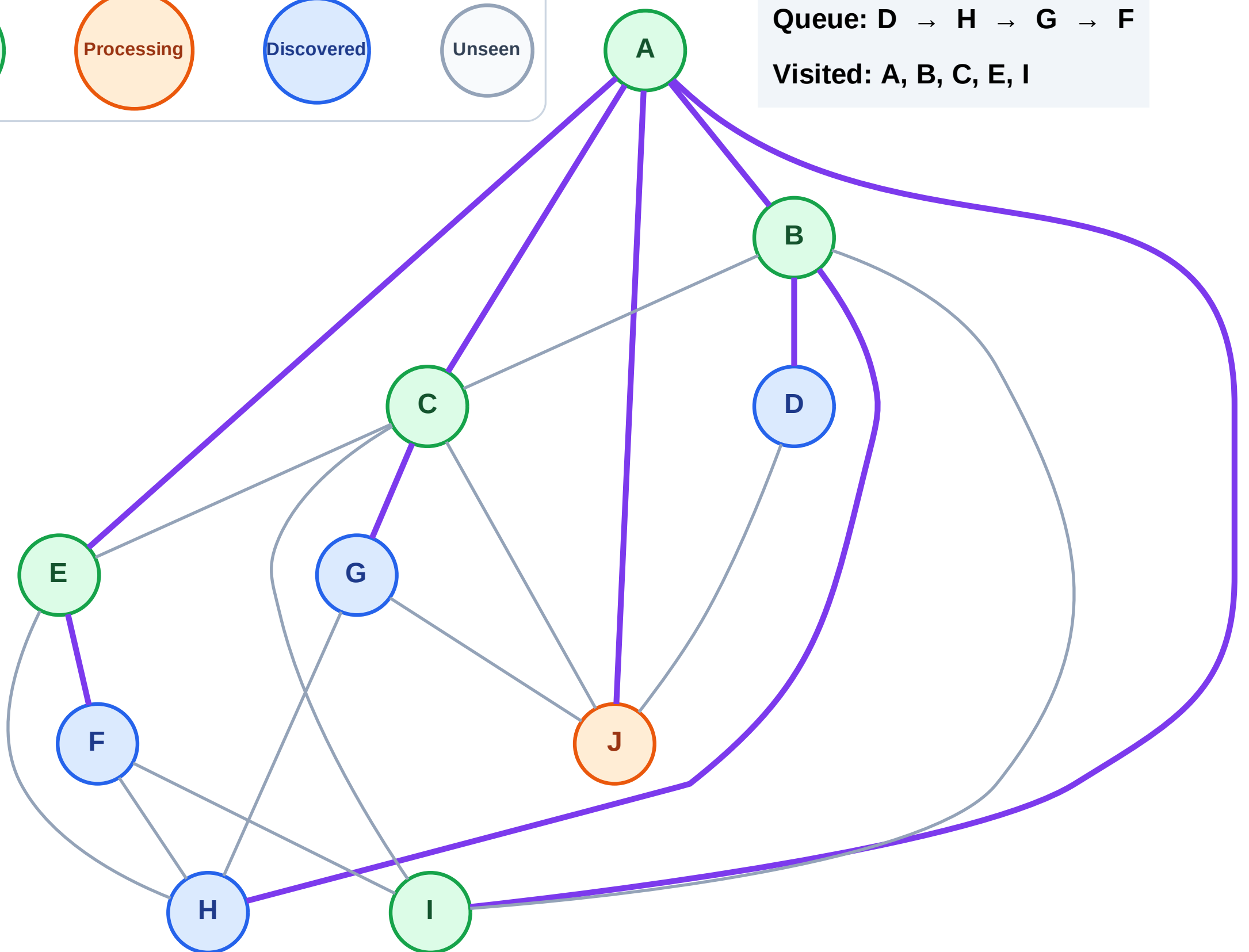
Discovered



Unseen

Queue: D → H → G → F

Visited: A, B, C, E, I



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

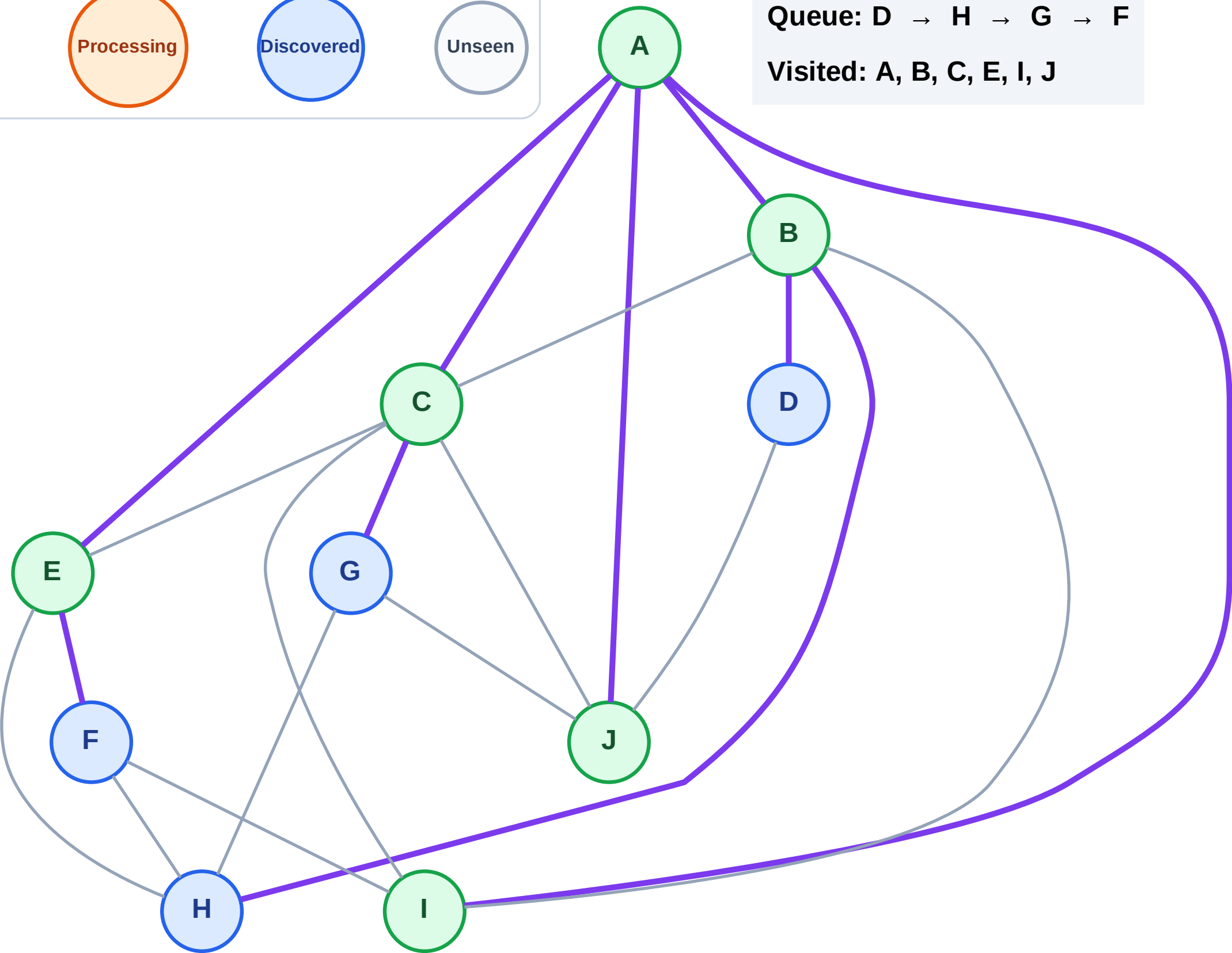
Processing

Discovered

Unseen

Queue: D → H → G → F

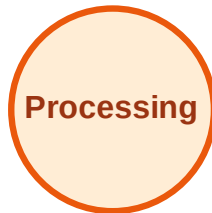
Visited: A, B, C, E, I, J



BFS Traversal

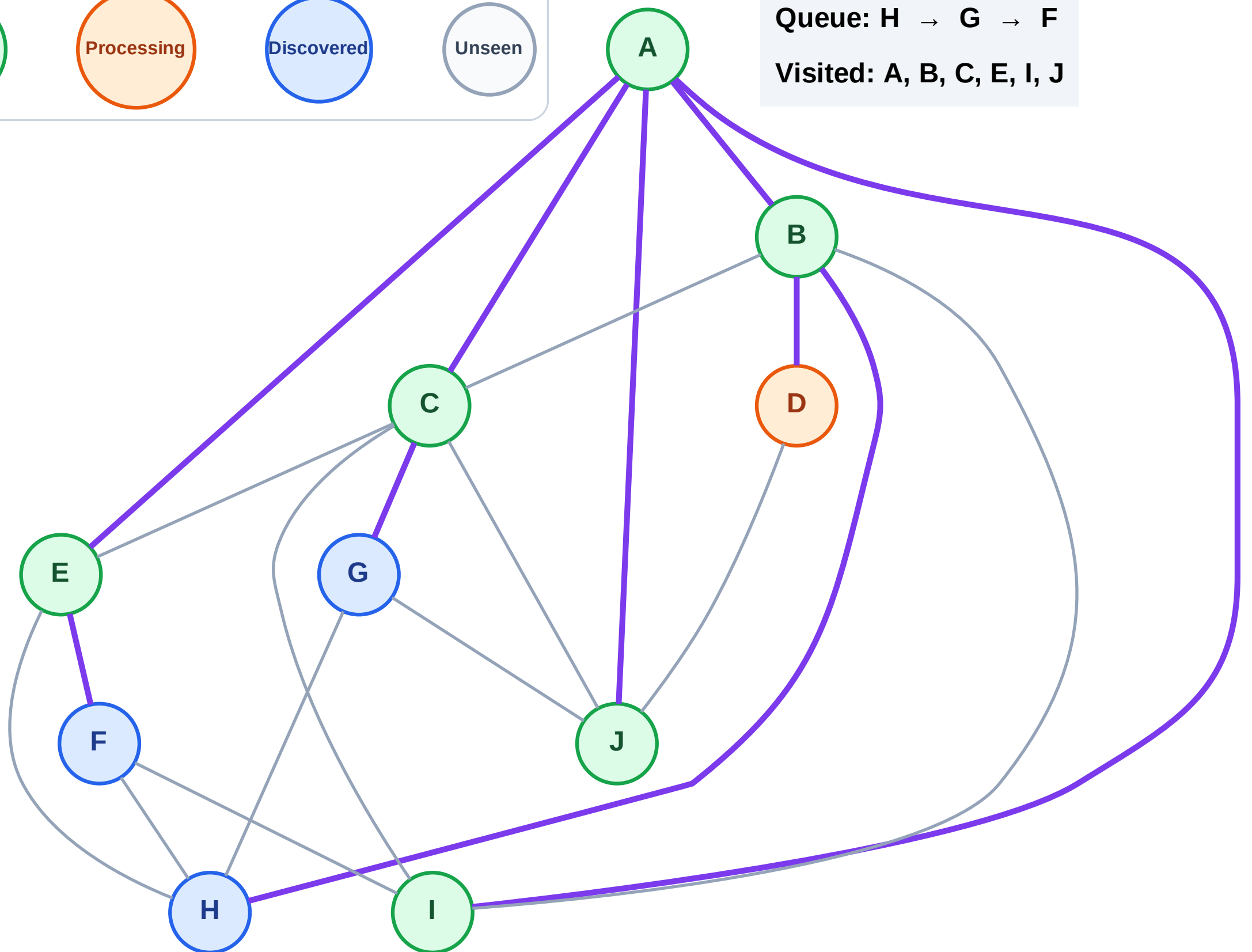
Step 14: Process node D

Legend



Queue: H → G → F

Visited: A, B, C, E, I, J



BFS Traversal

Step 15: No new neighbors from D

Legend

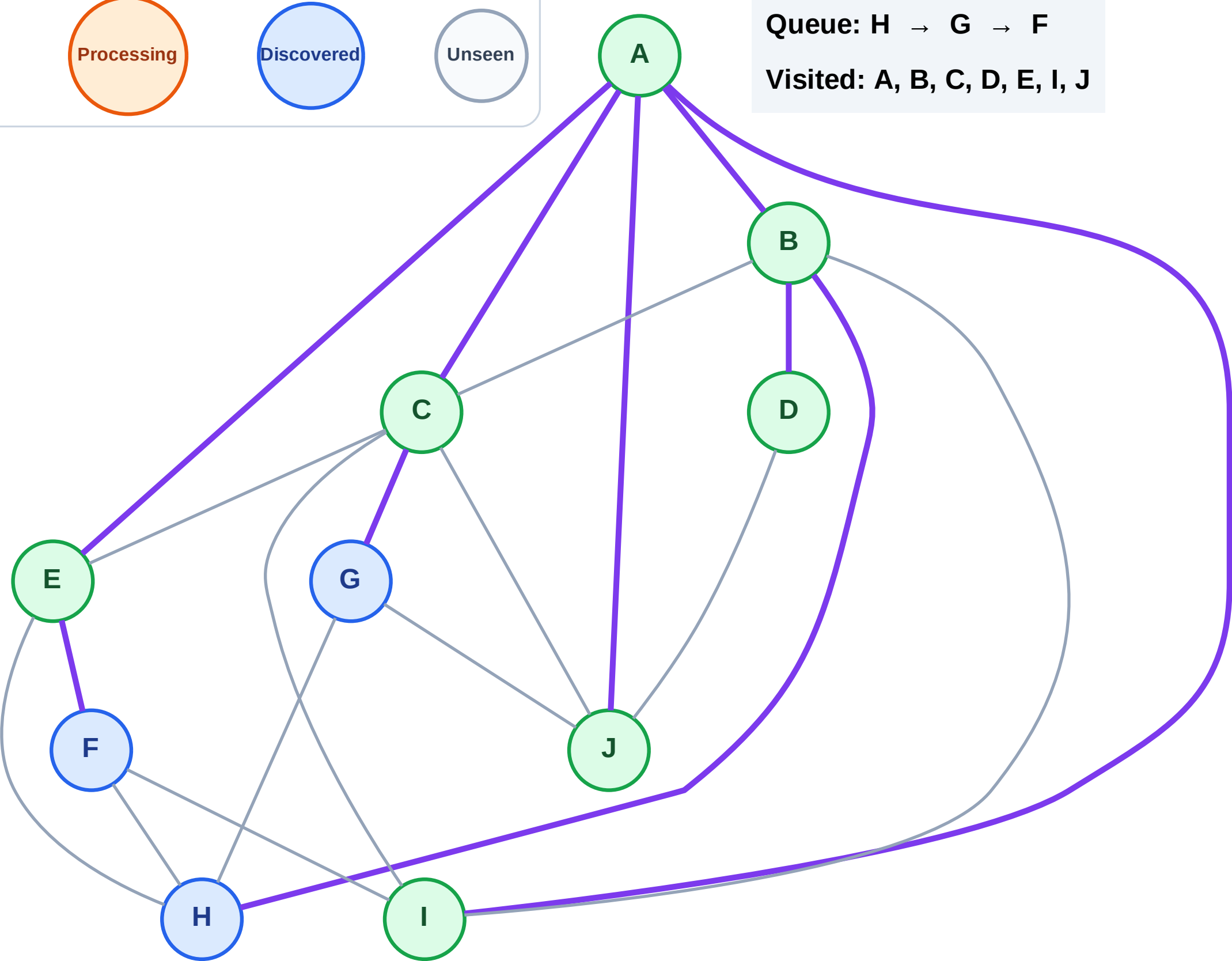
Complete

Processing

Discovered

Unseen

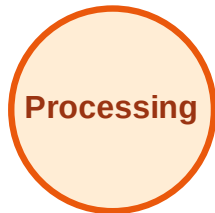
Queue: H → G → F
Visited: A, B, C, D, E, I, J



BFS Traversal

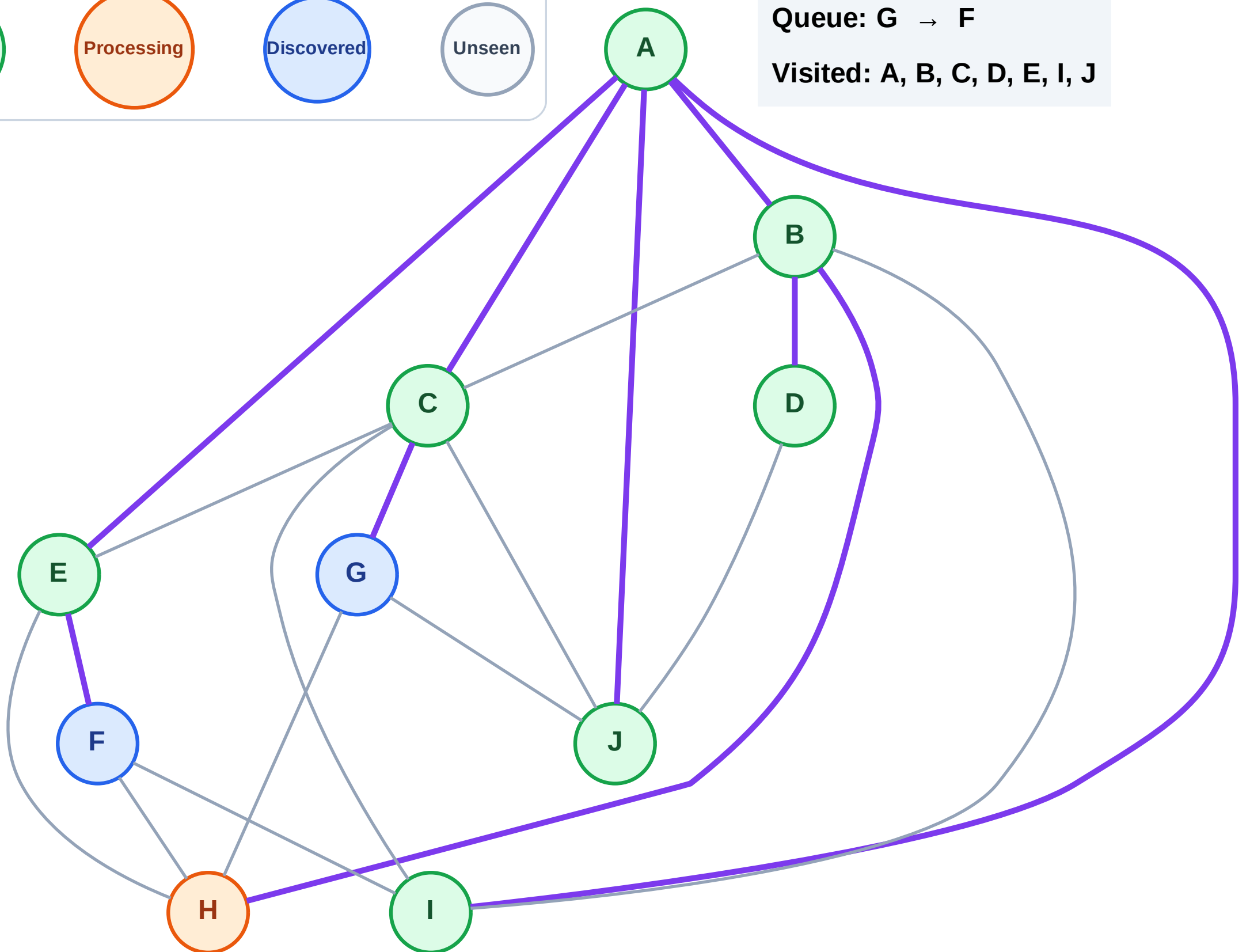
Step 16: Process node H

Legend



Queue: G → F

Visited: A, B, C, D, E, I, J



BFS Traversal

Step 17: No new neighbors from H

Legend

Complete

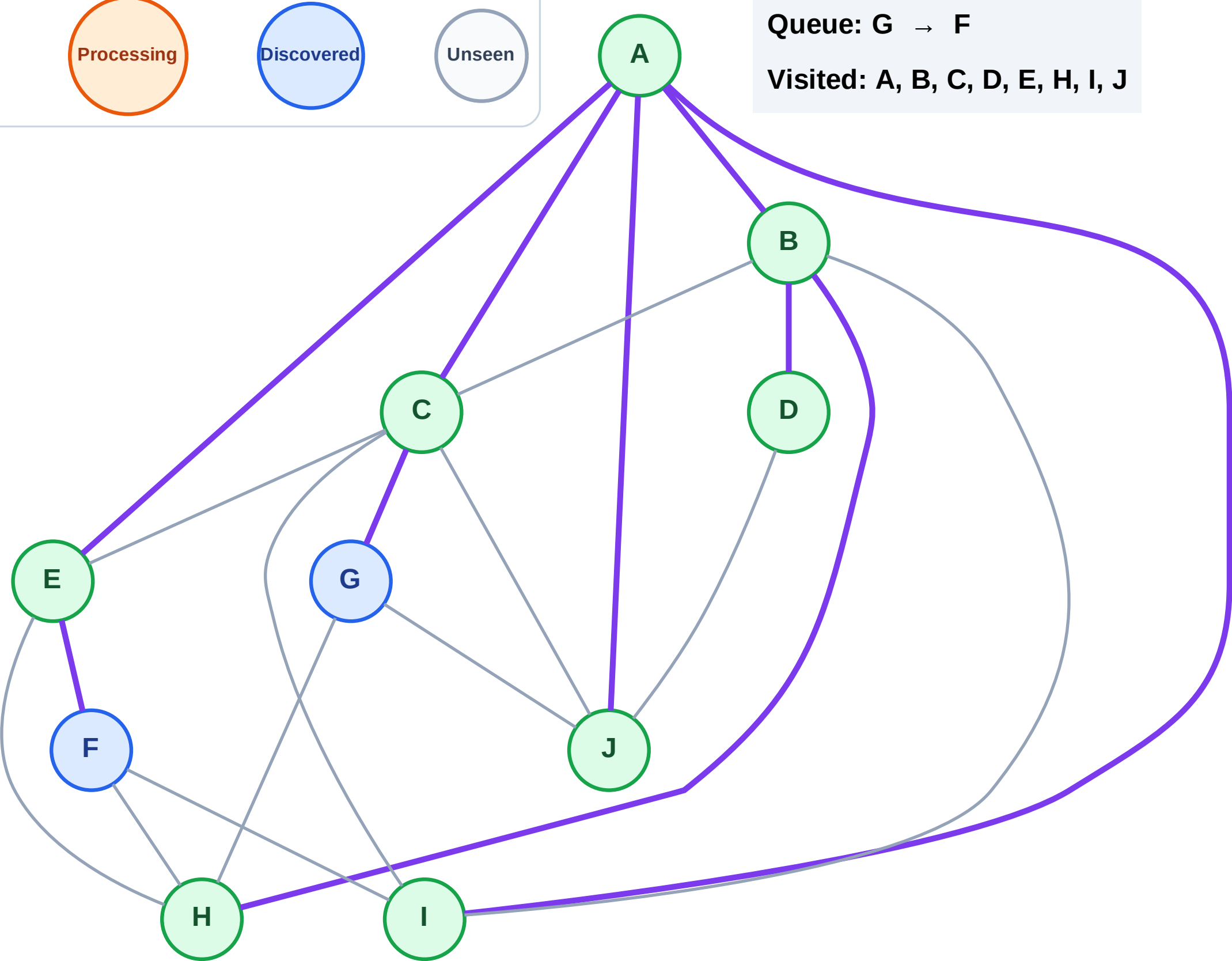
Processing

Discovered

Unseen

Queue: G → F

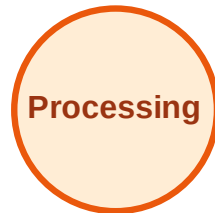
Visited: A, B, C, D, E, H, I, J



BFS Traversal

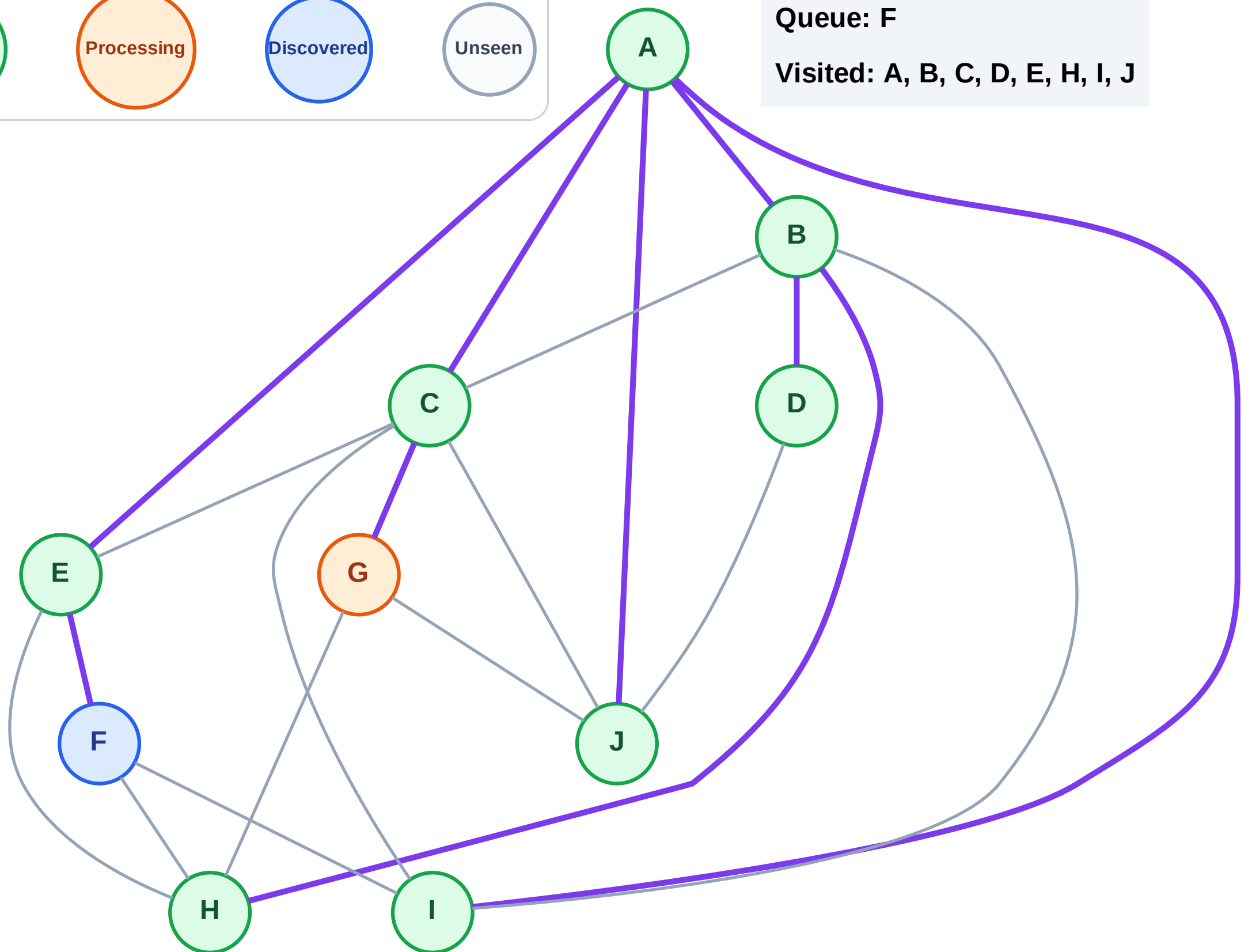
Step 18: Process node G

Legend



Queue: F

Visited: A, B, C, D, E, H, I, J



BFS Traversal

Step 19: No new neighbors from G

Legend

Complete

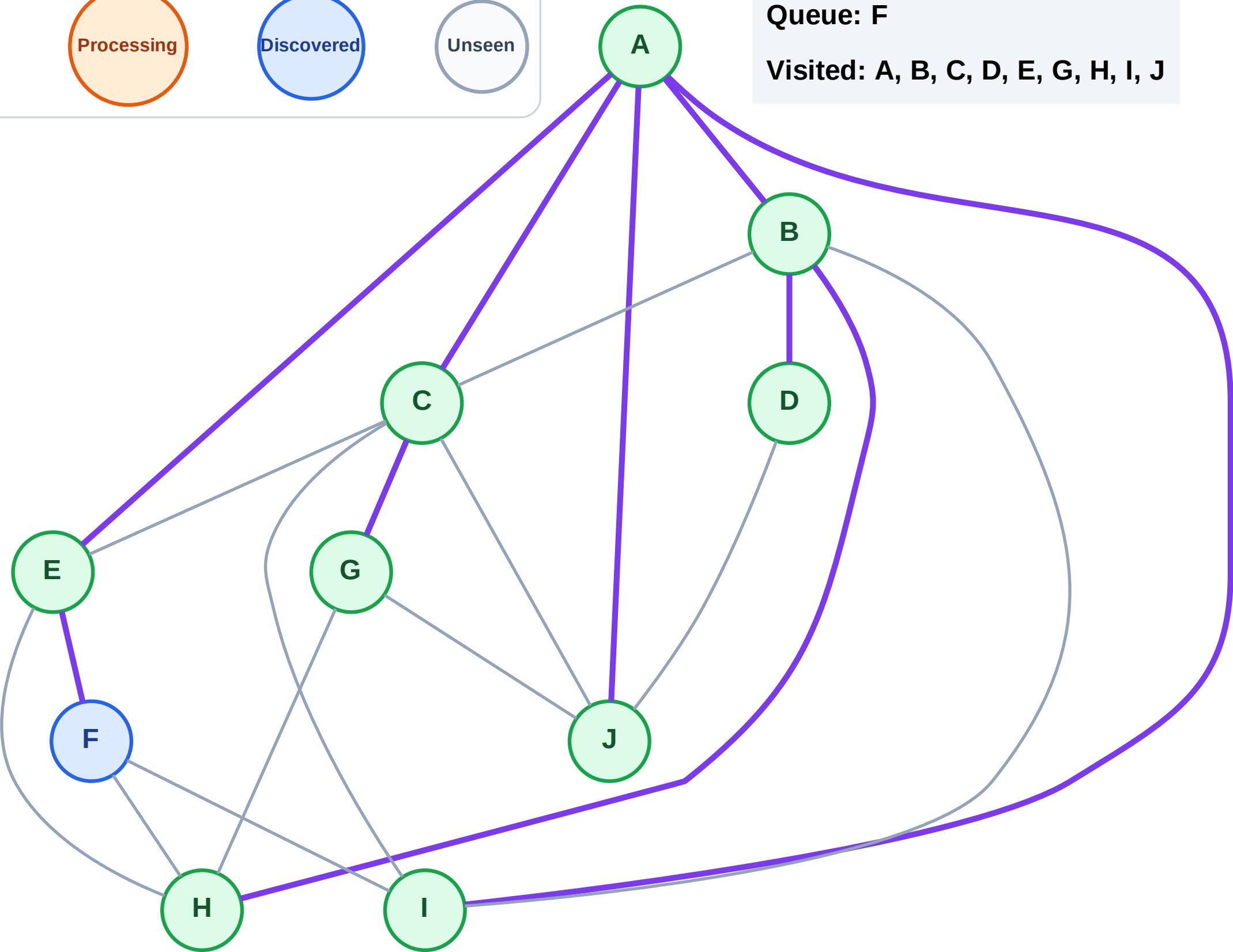
Processing

Discovered

Unseen

Queue: F

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

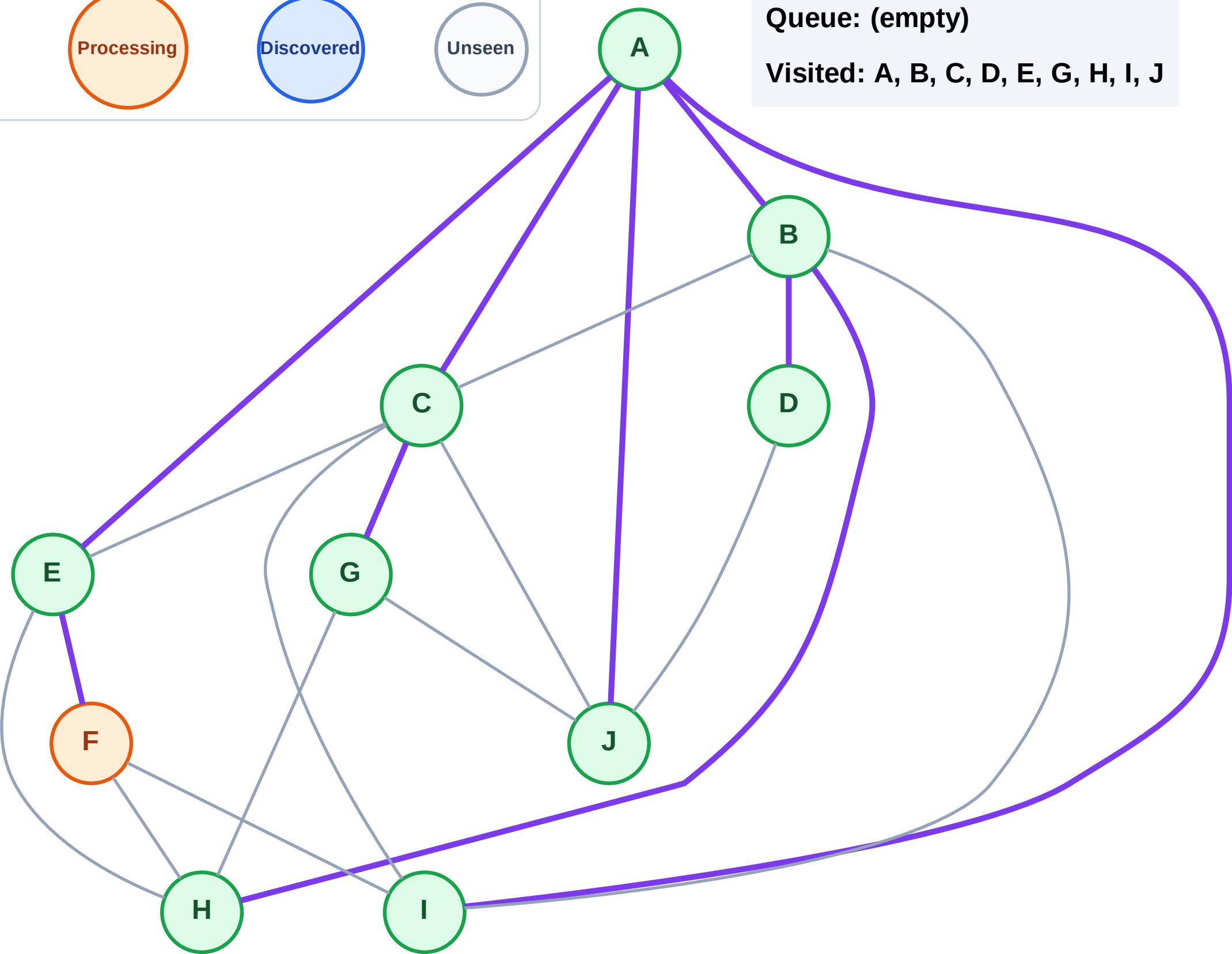
Step 20: Process node F

Legend



Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

Step 21: No new neighbors from F

Legend

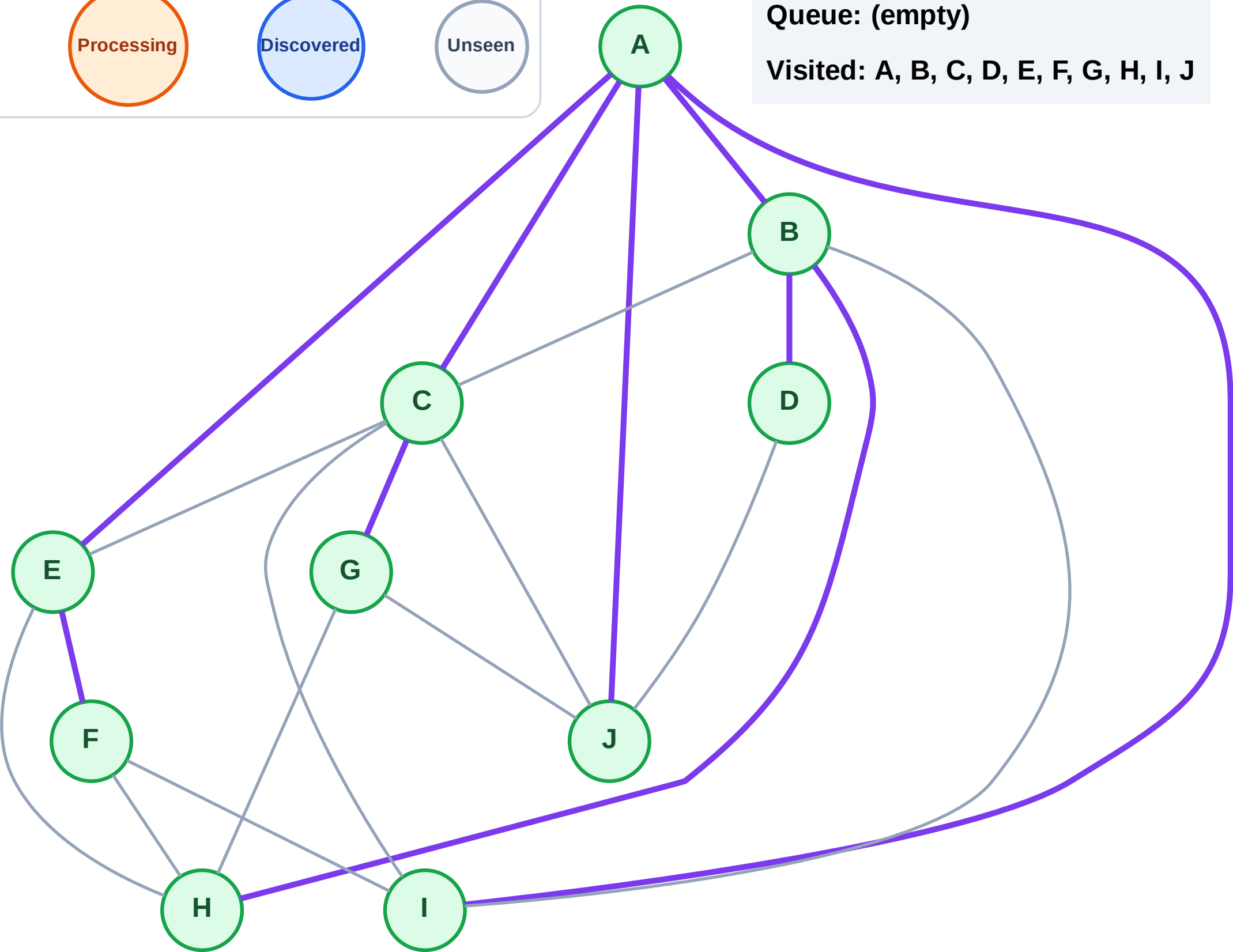
Complete

Processing

Discovered

Unseen

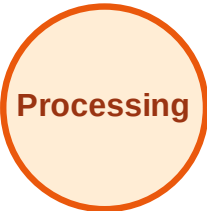
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

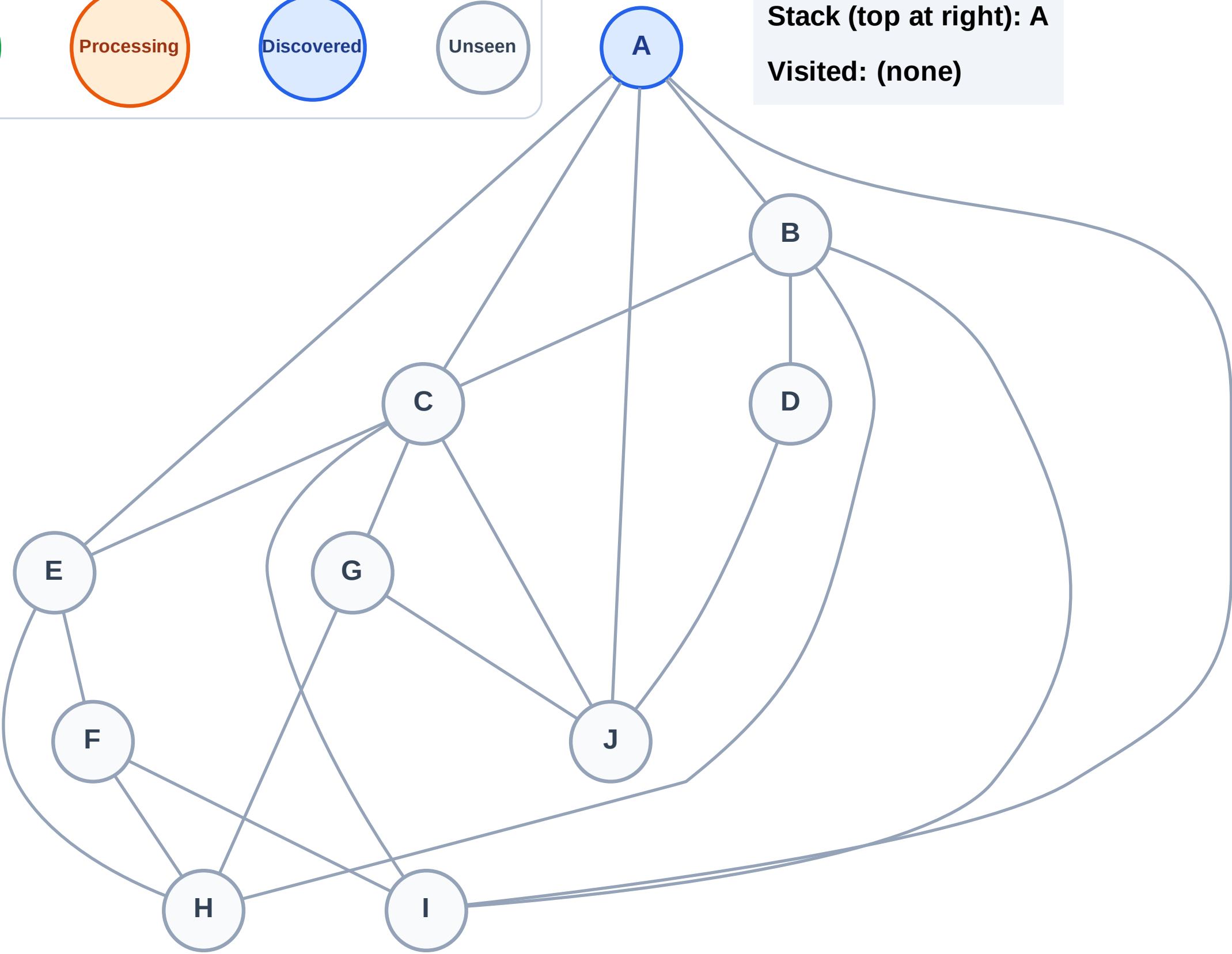
Step 1: Start at A

Legend



Stack (top at right): A

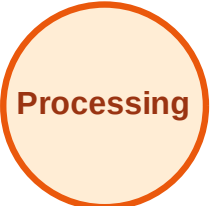
Visited: (none)



DFS Traversal

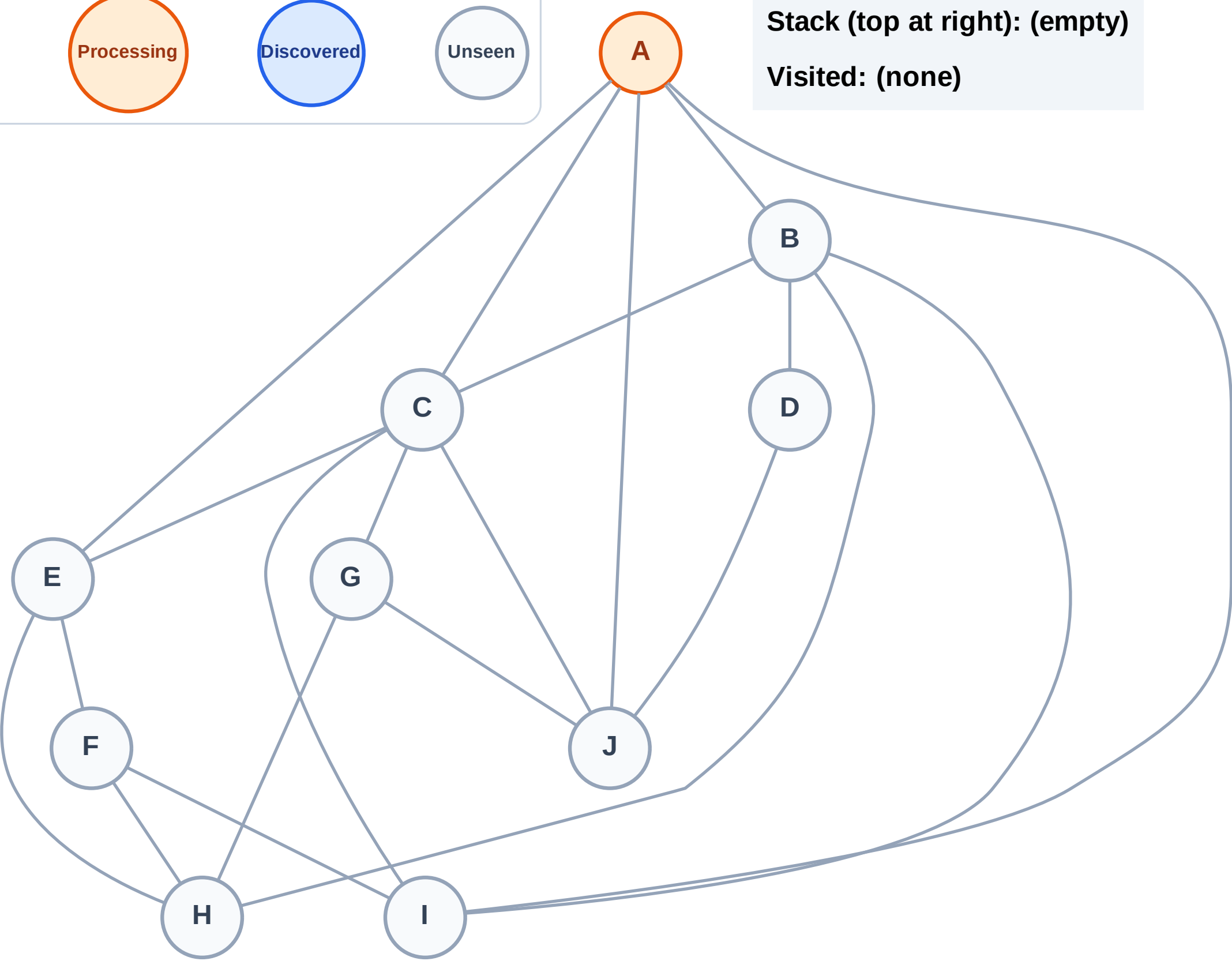
Step 2: Process node A

Legend



Stack (top at right): (empty)

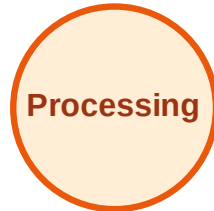
Visited: (none)



DFS Traversal

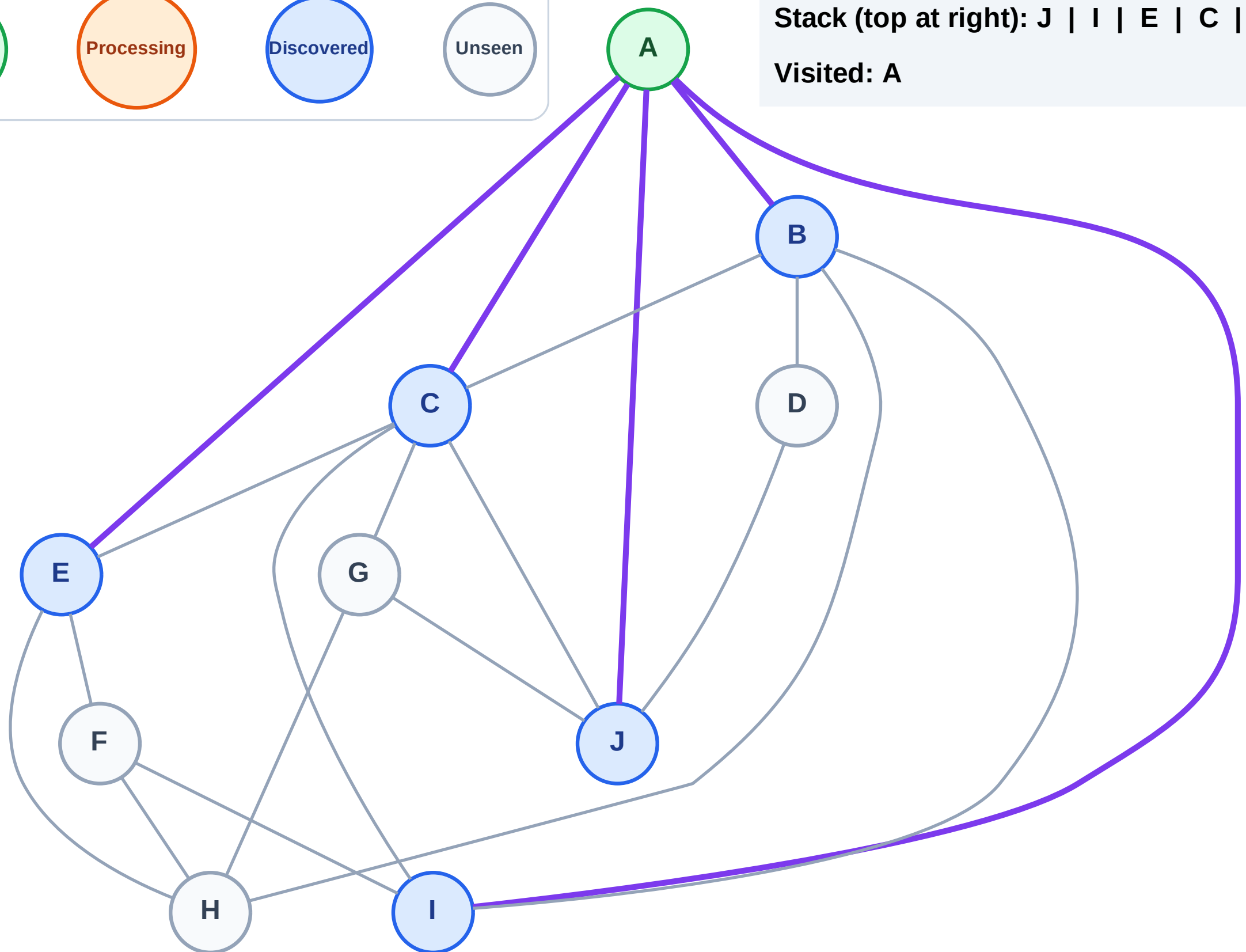
Step 3: Discover J, I, E, C, B from A

Legend



Stack (top at right): J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

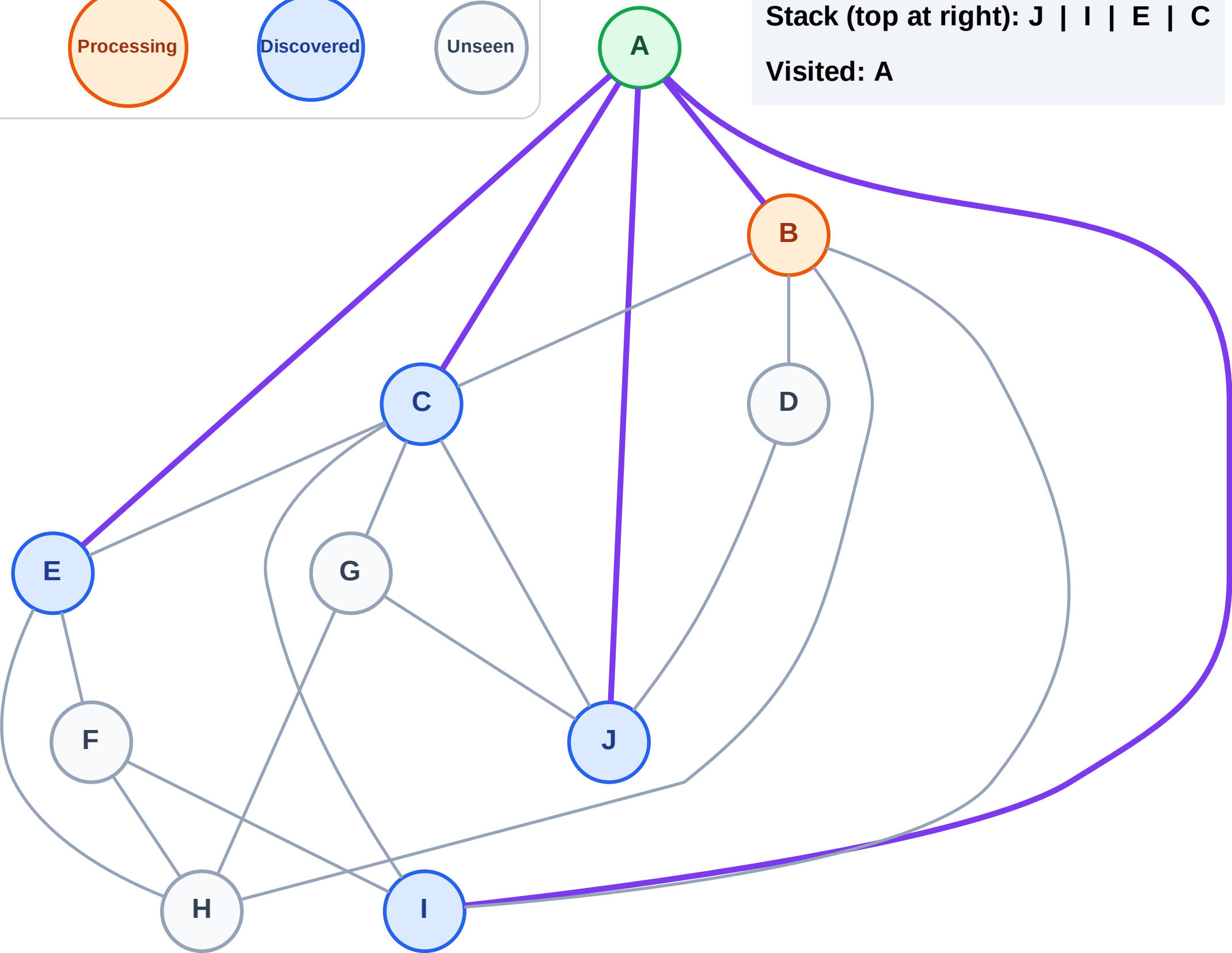
Processing

Discovered

Unseen

Stack (top at right): J | I | E | C

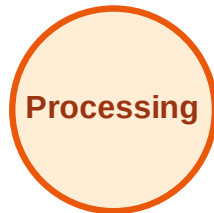
Visited: A



DFS Traversal

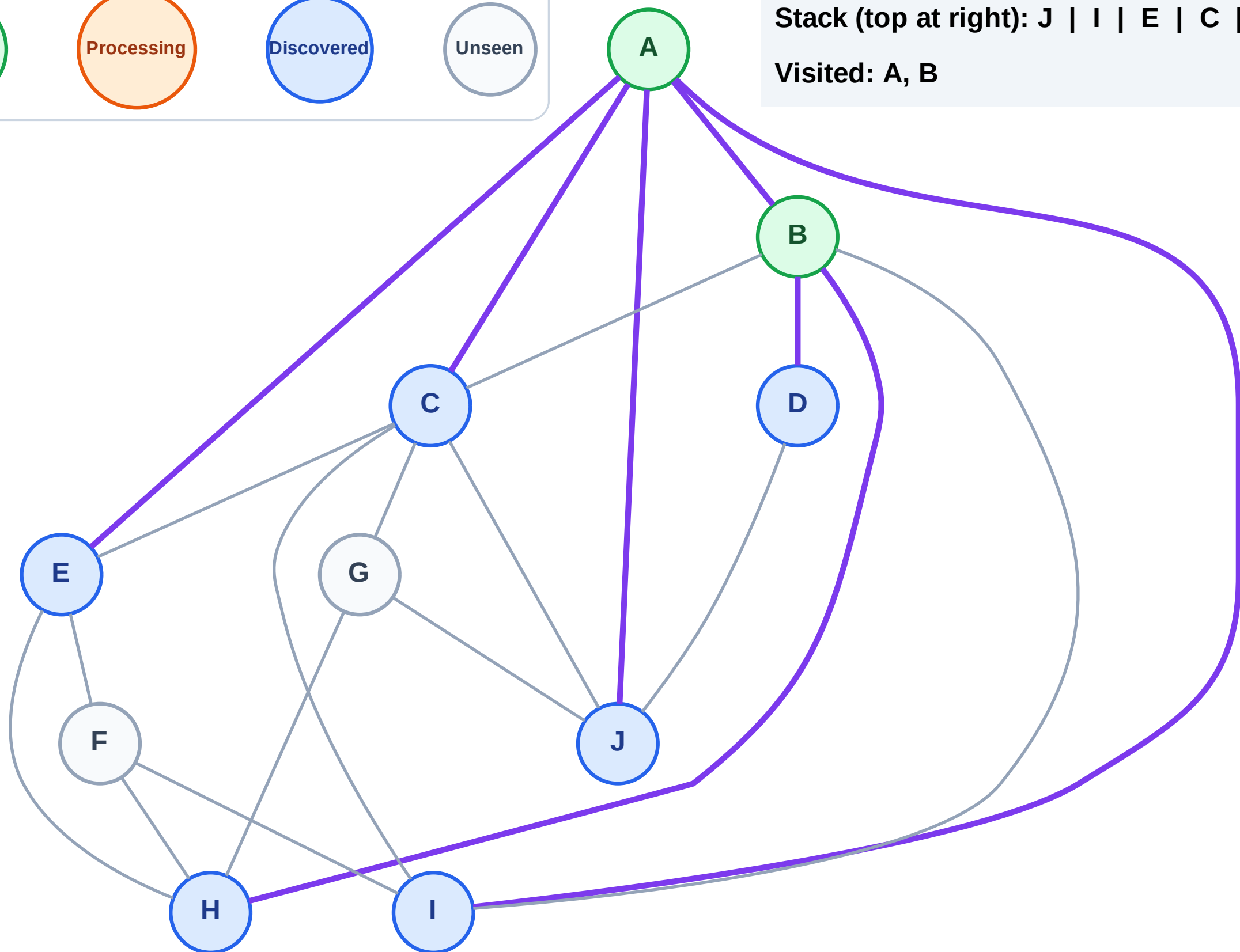
Step 5: Discover H, D from B

Legend



Stack (top at right): J | I | E | C | H | D

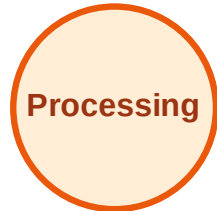
Visited: A, B



DFS Traversal

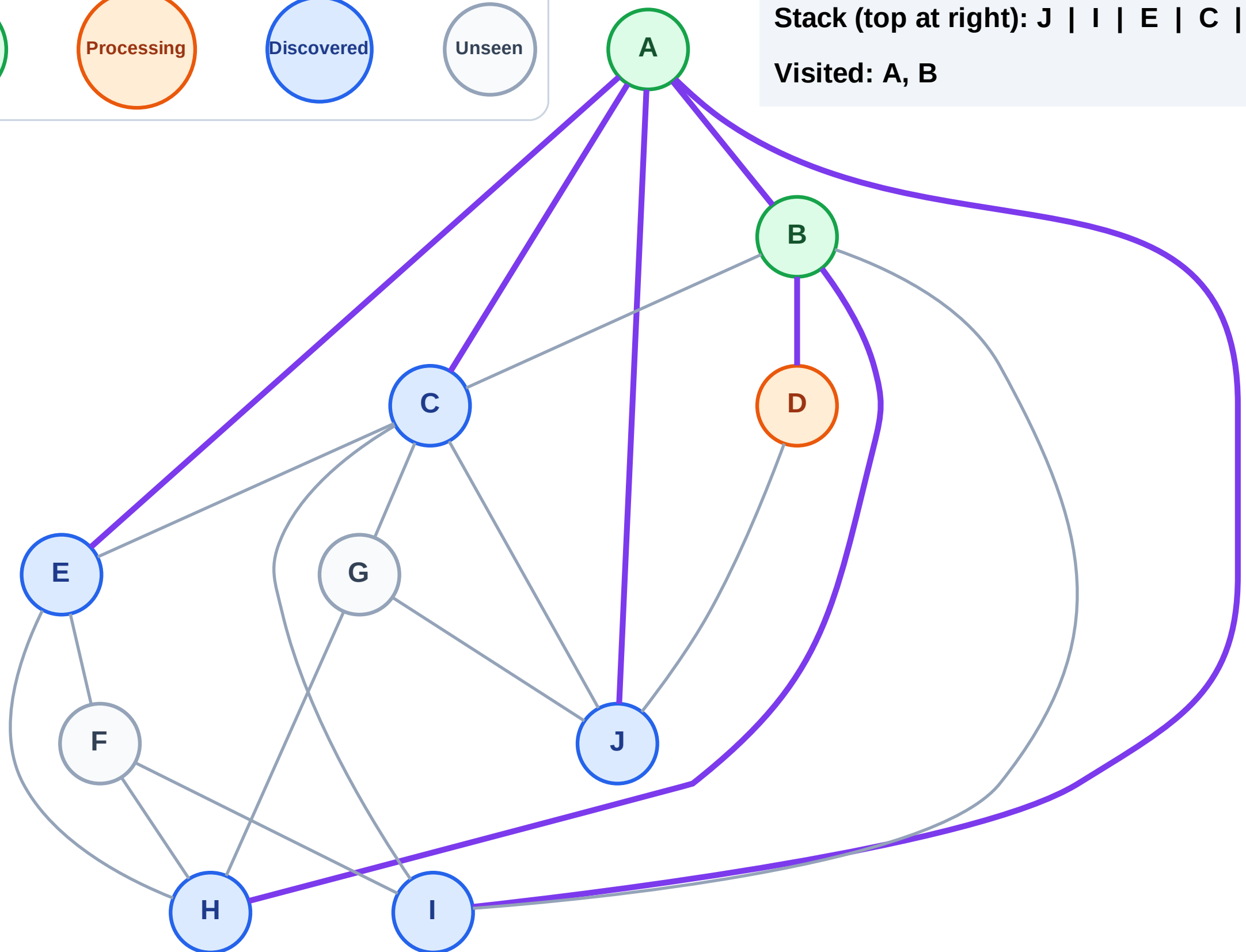
Step 6: Process node D

Legend



Stack (top at right): J | I | E | C | H

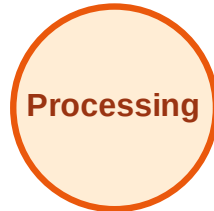
Visited: A, B



DFS Traversal

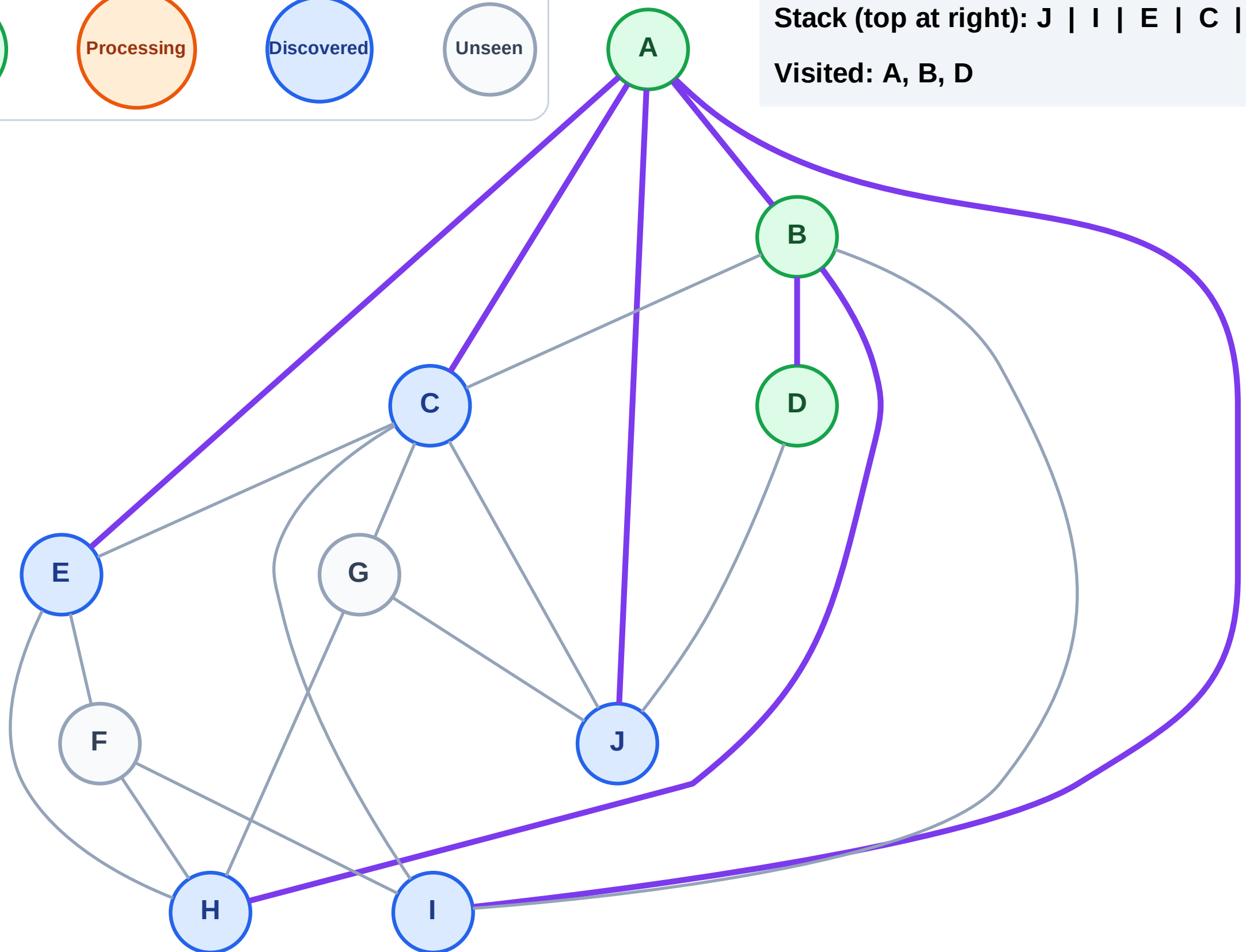
Step 7: No new neighbors from D

Legend



Stack (top at right): J | I | E | C | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

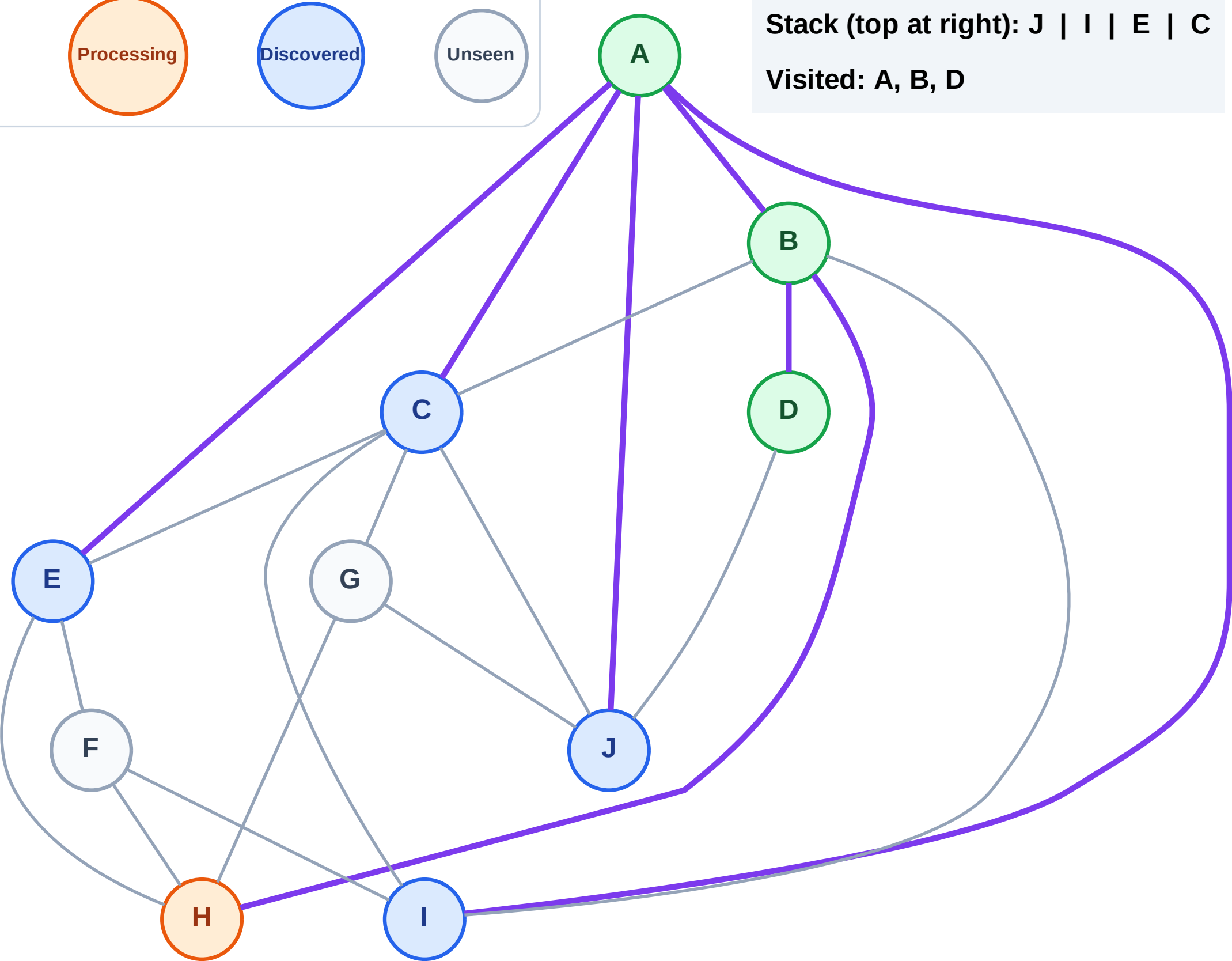
Complete

Processing

Discovered

Unseen

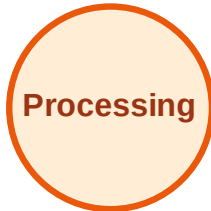
Stack (top at right): J | I | E | C
Visited: A, B, D



DFS Traversal

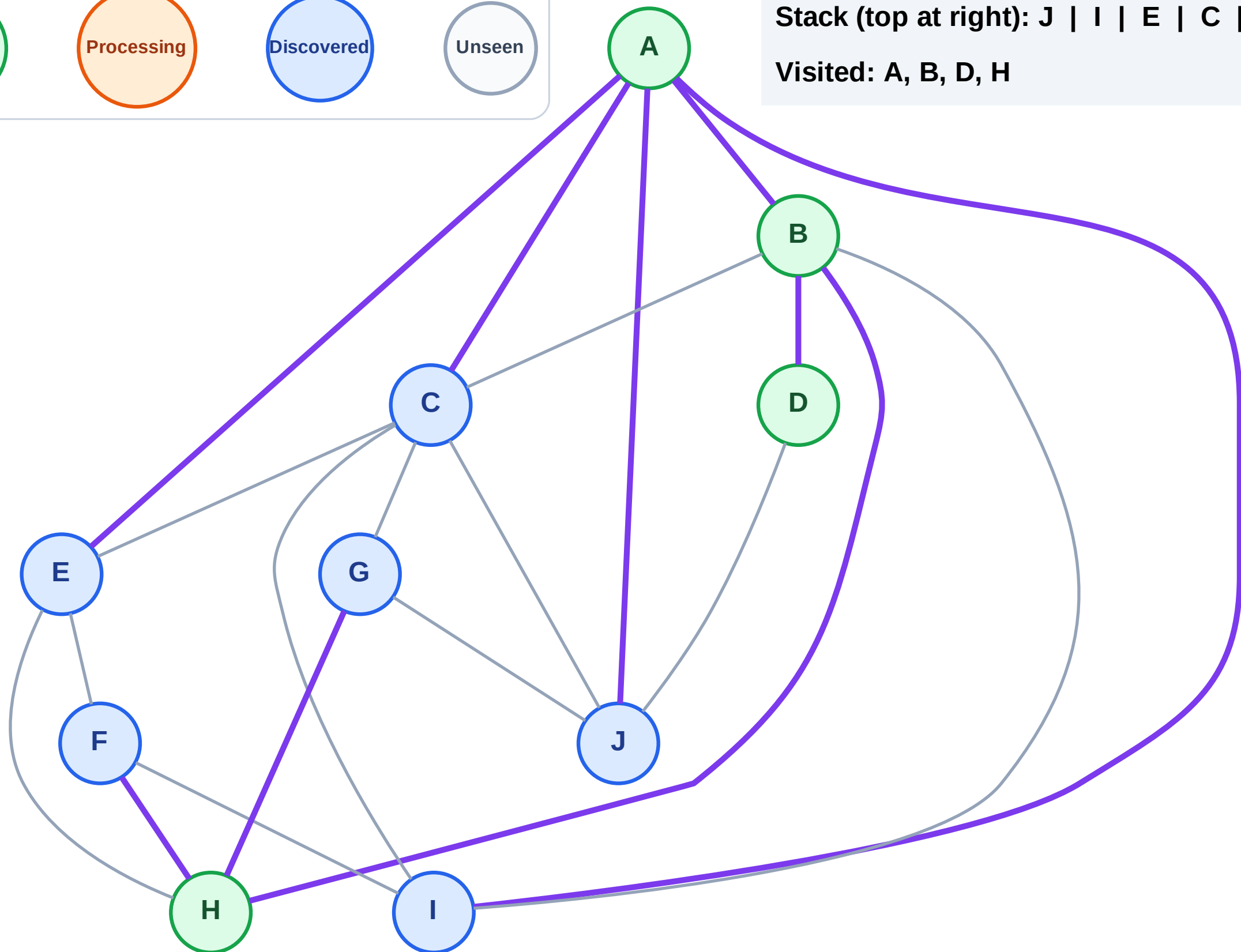
Step 9: Discover G, F from H

Legend



Stack (top at right): J | I | E | C | G | F

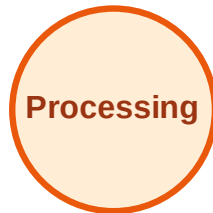
Visited: A, B, D, H



DFS Traversal

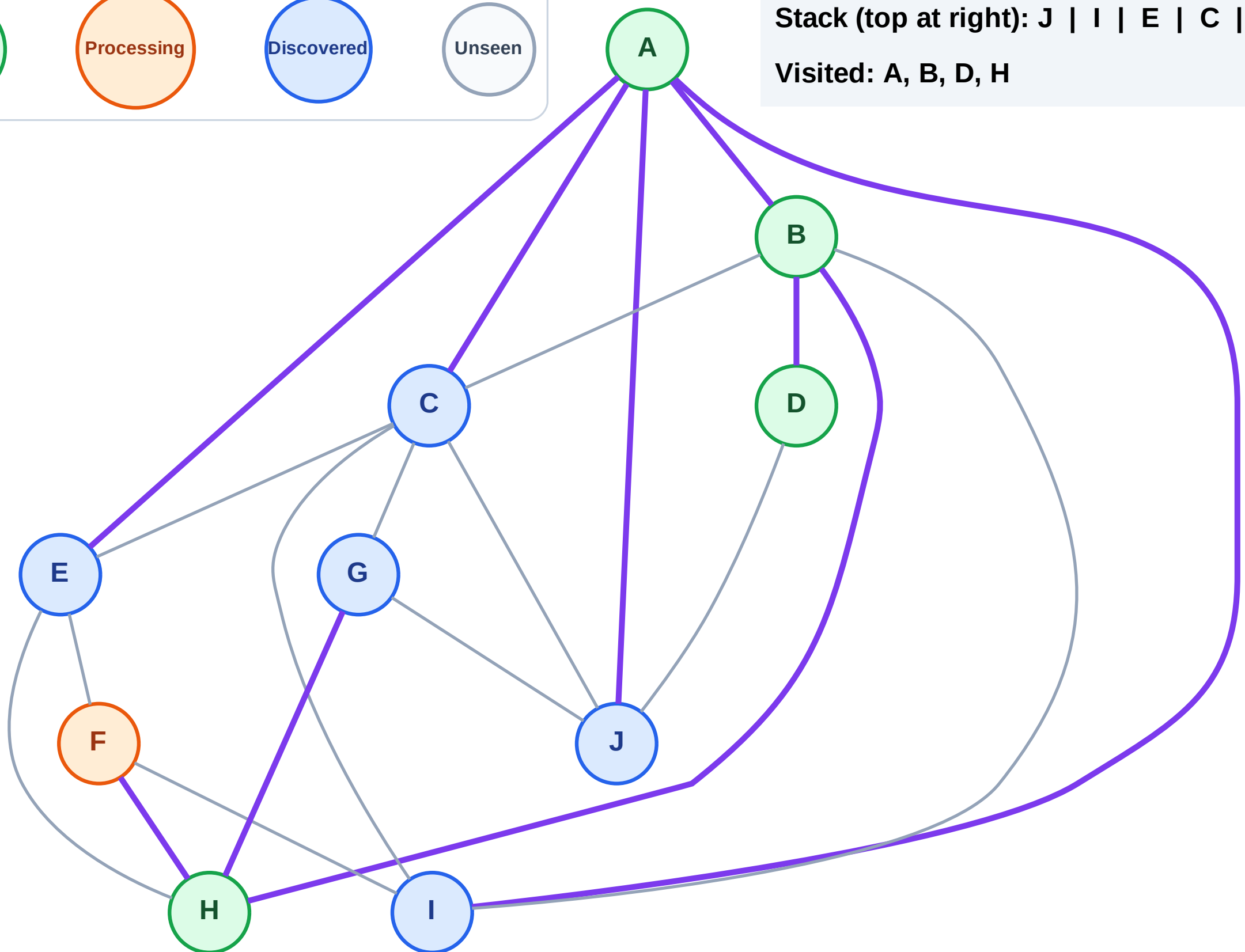
Step 10: Process node F

Legend



Stack (top at right): J | I | E | C | G

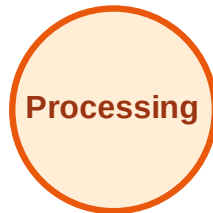
Visited: A, B, D, H



DFS Traversal

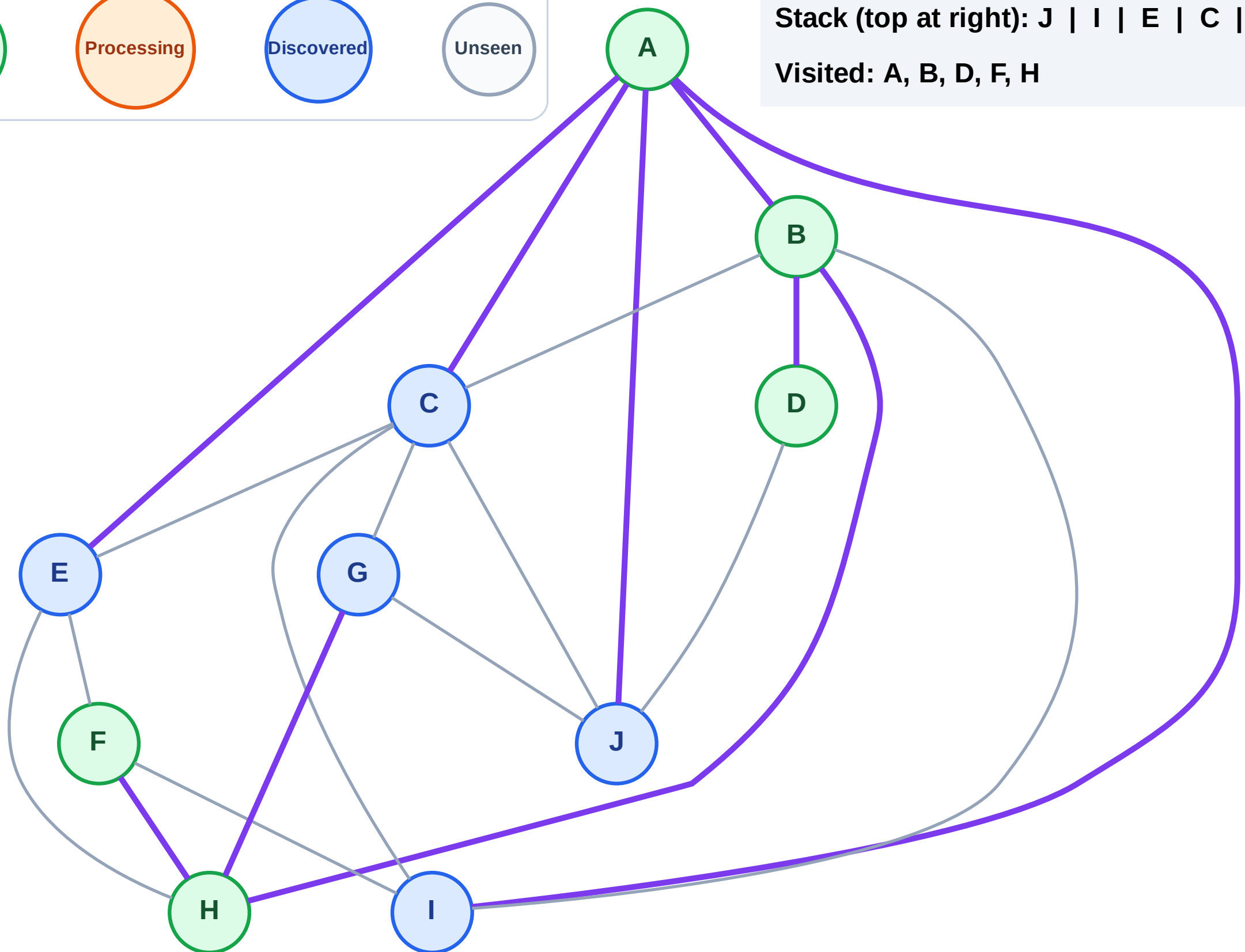
Step 11: No new neighbors from F

Legend



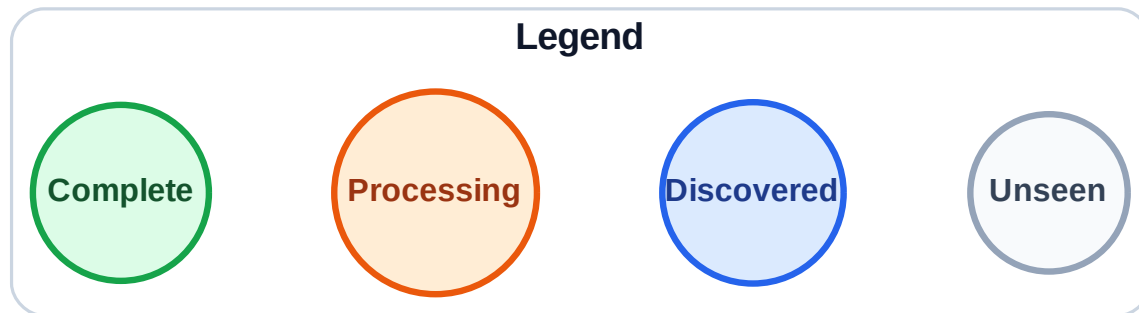
Stack (top at right): J | I | E | C | G

Visited: A, B, D, F, H



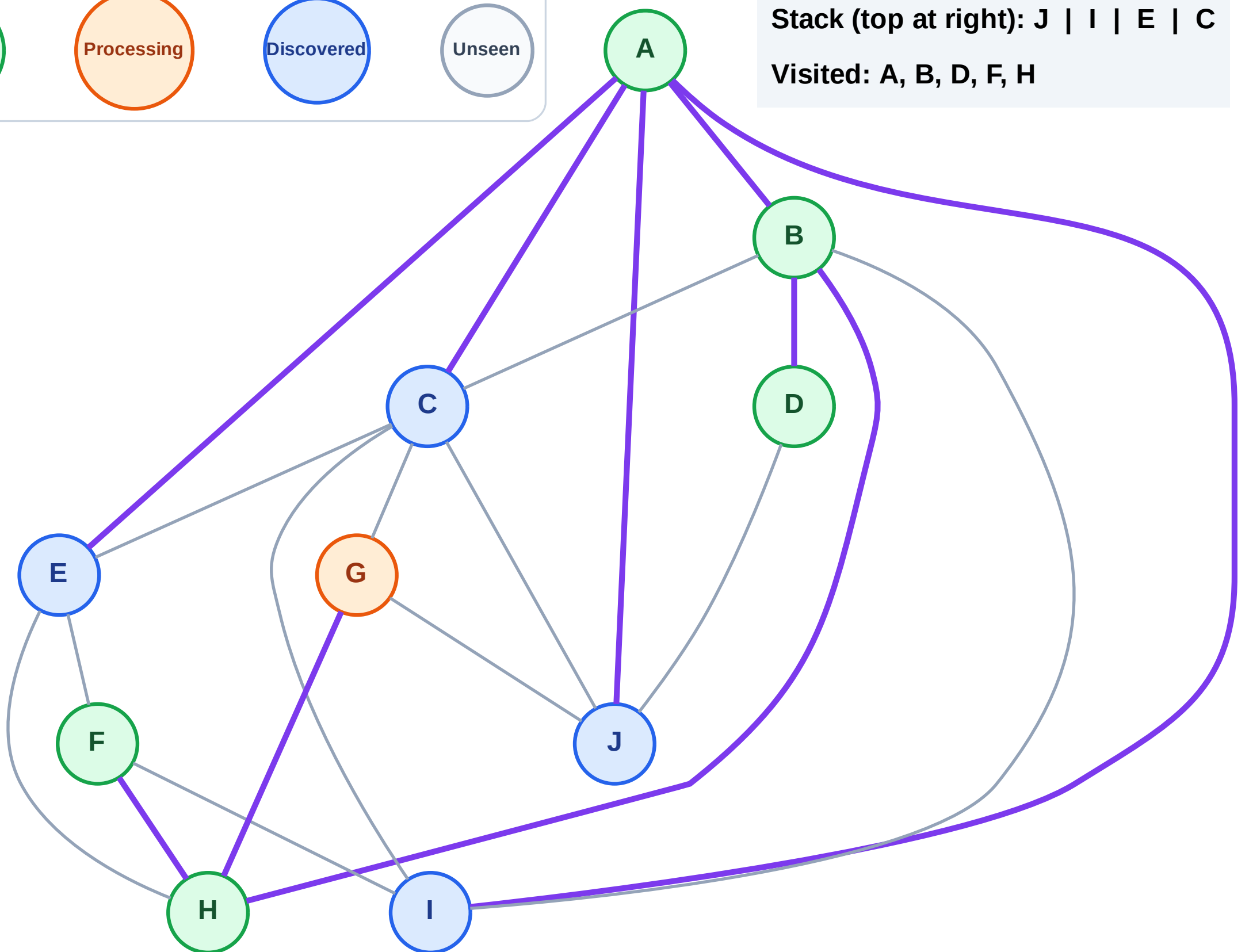
Step 12: Process node G

Step 12: Process node G



Stack (top at right): J | I | E | C

Visited: A, B, D, F, H



DFS Traversal

Step 13: No new neighbors from G

Legend

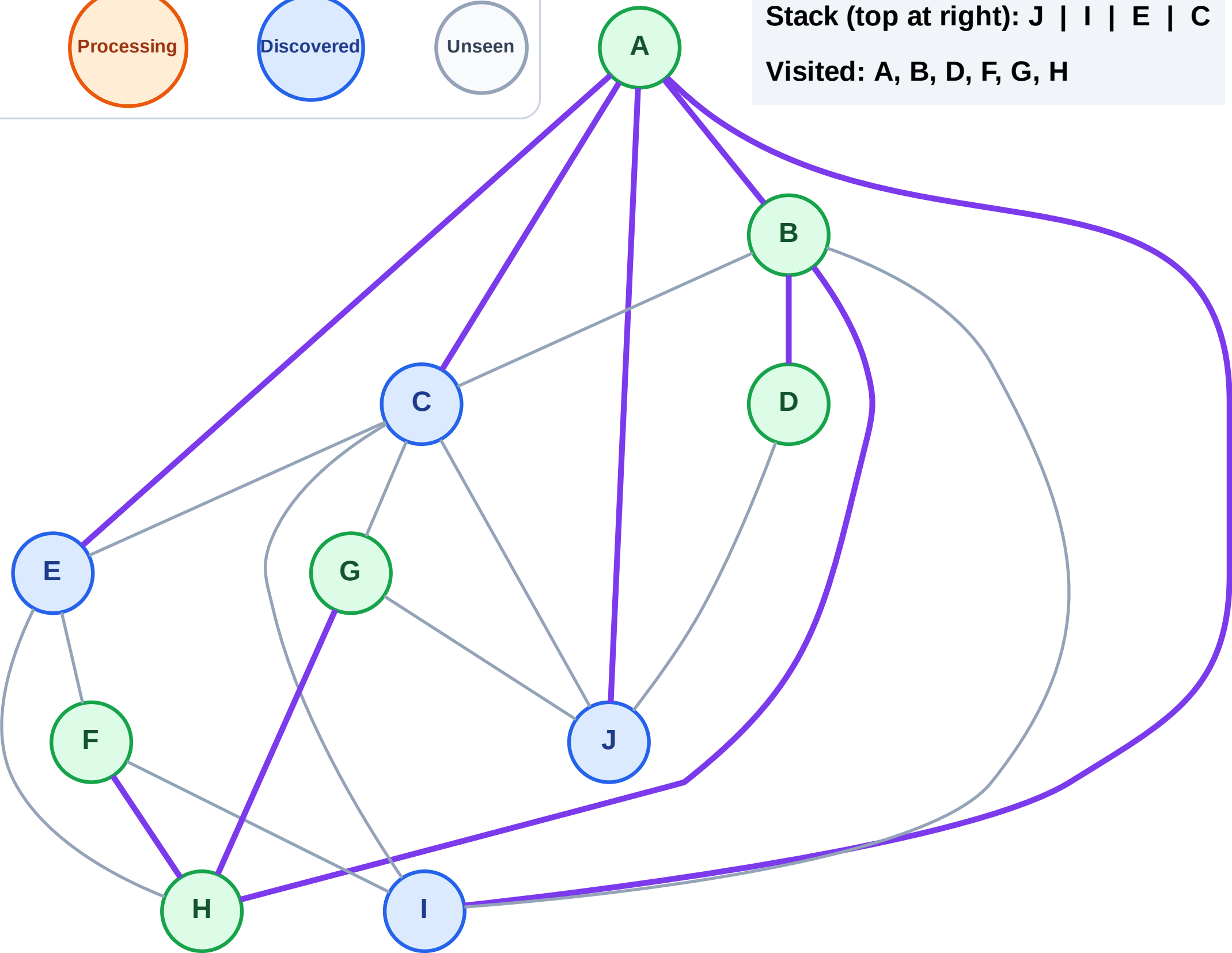
Complete

Processing

Discovered

Unseen

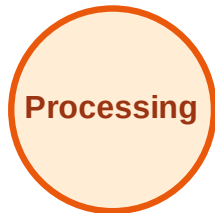
Stack (top at right): J | I | E | C
Visited: A, B, D, F, G, H



DFS Traversal

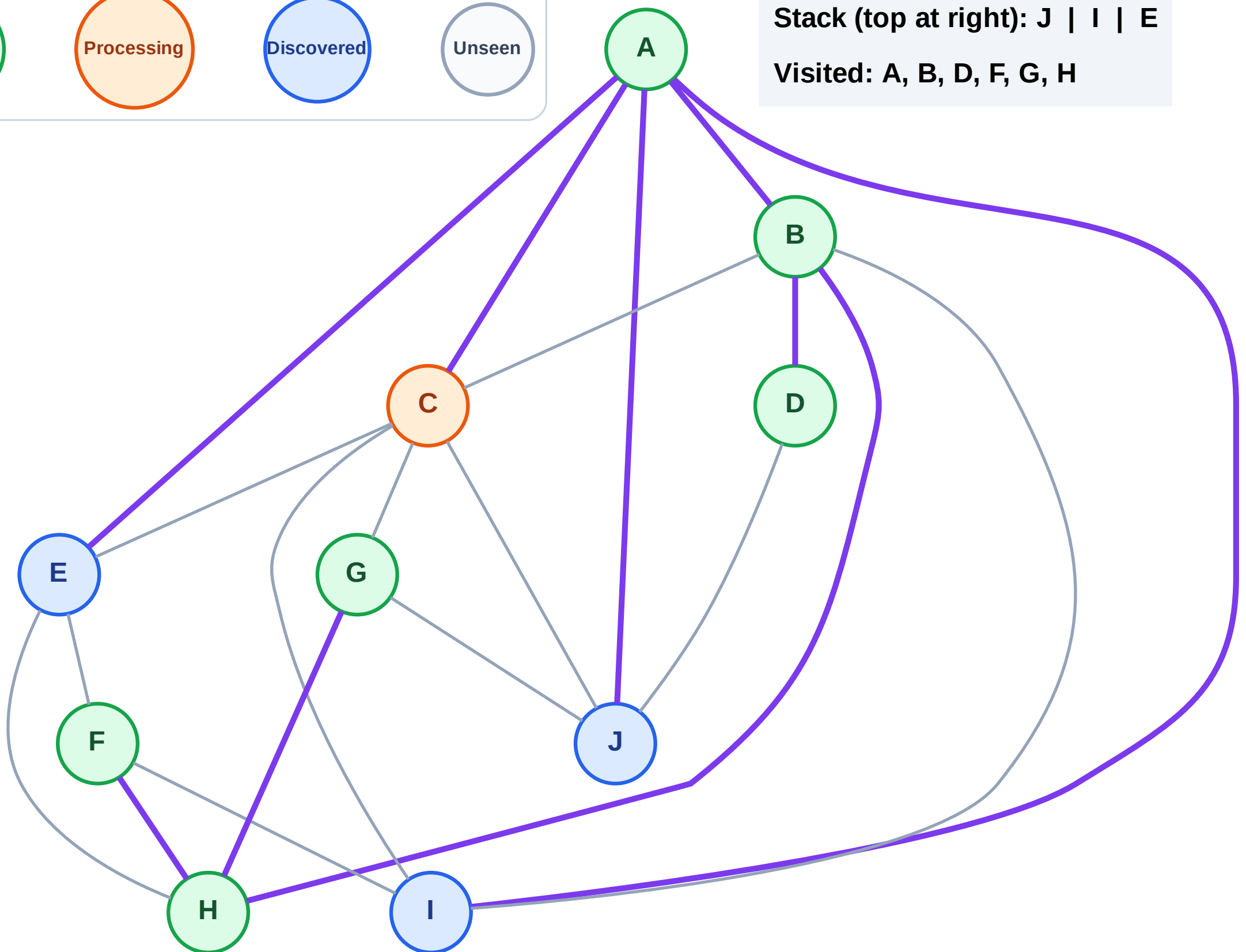
Step 14: Process node C

Legend



Stack (top at right): J | I | E

Visited: A, B, D, F, G, H



DFS Traversal

Step 15: No new neighbors from C

Legend

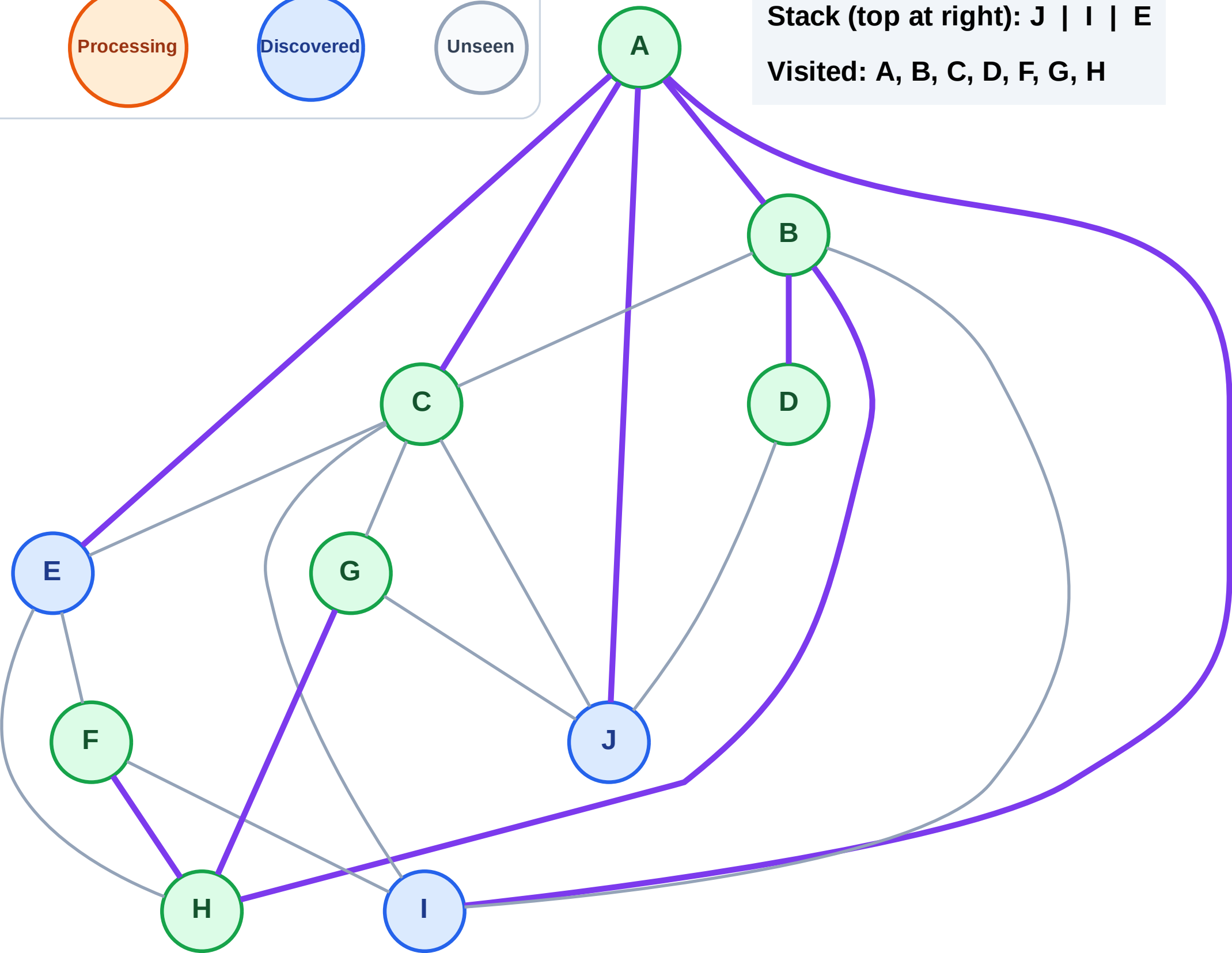
Complete

Processing

Discovered

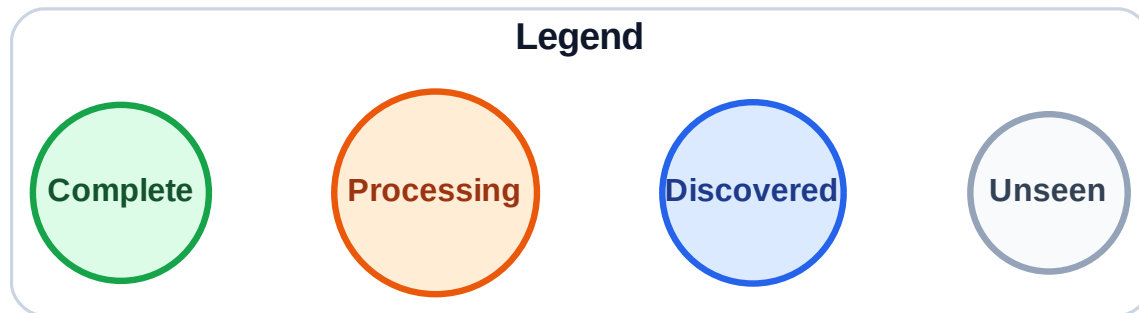
Unseen

Stack (top at right): J | I | E
Visited: A, B, C, D, F, G, H



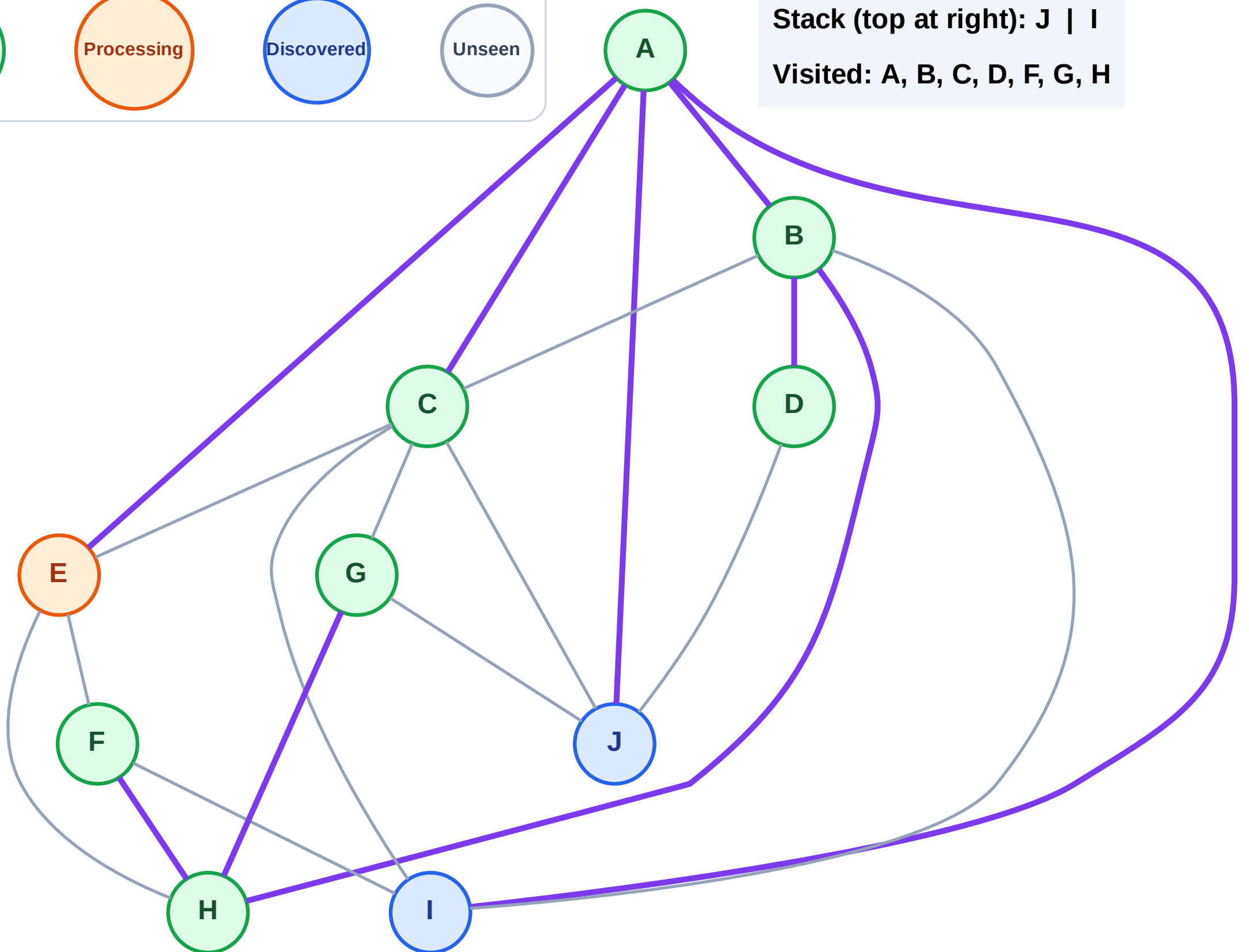
Step 16: Process node E

Step 16: Process node E



Stack (top at right): J | I

Visited: A, B, C, D, F, G, H



DFS Traversal

Step 17: No new neighbors from E

Legend

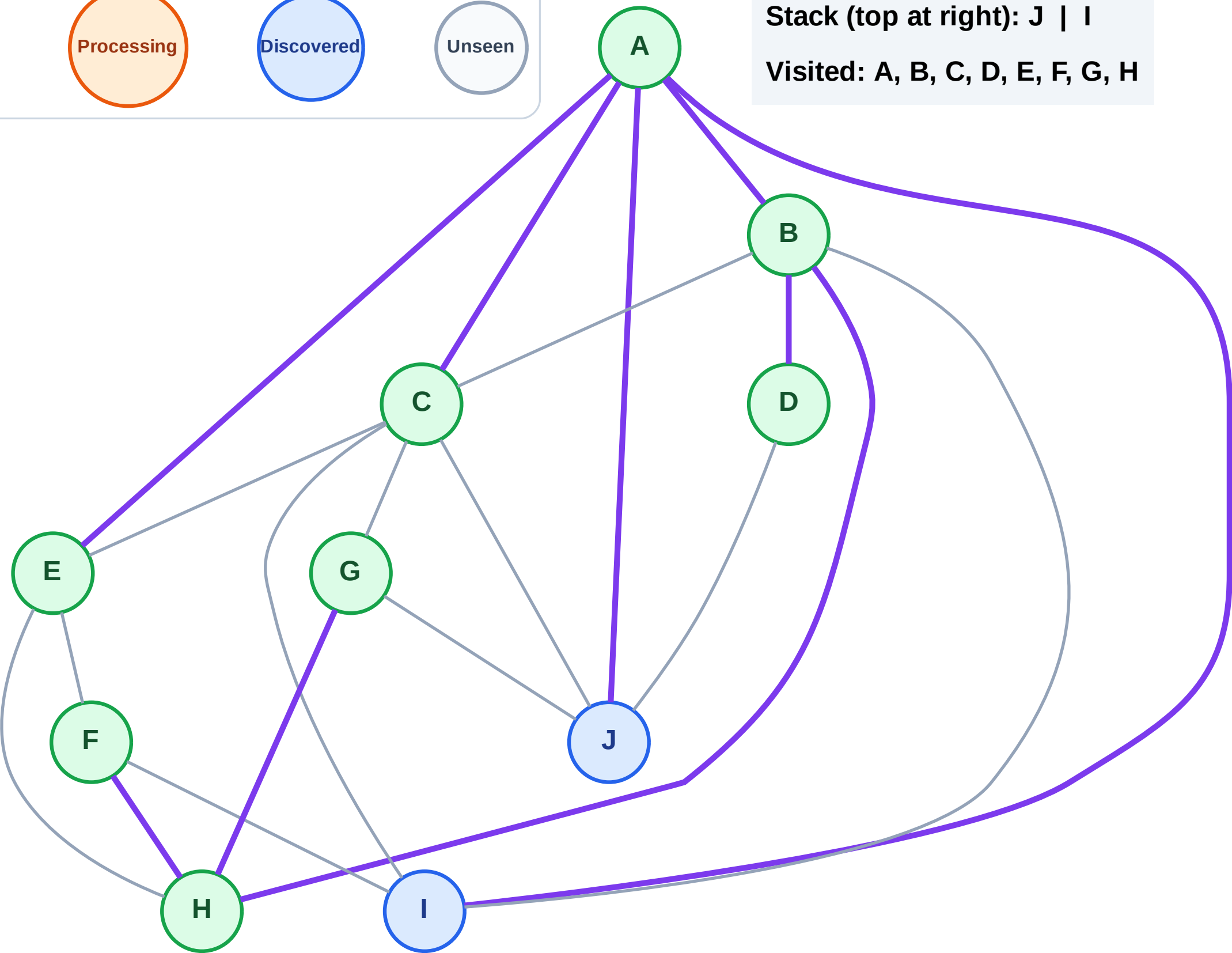
Complete

Processing

Discovered

Unseen

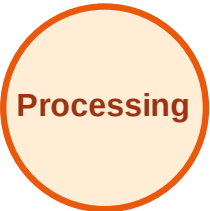
Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

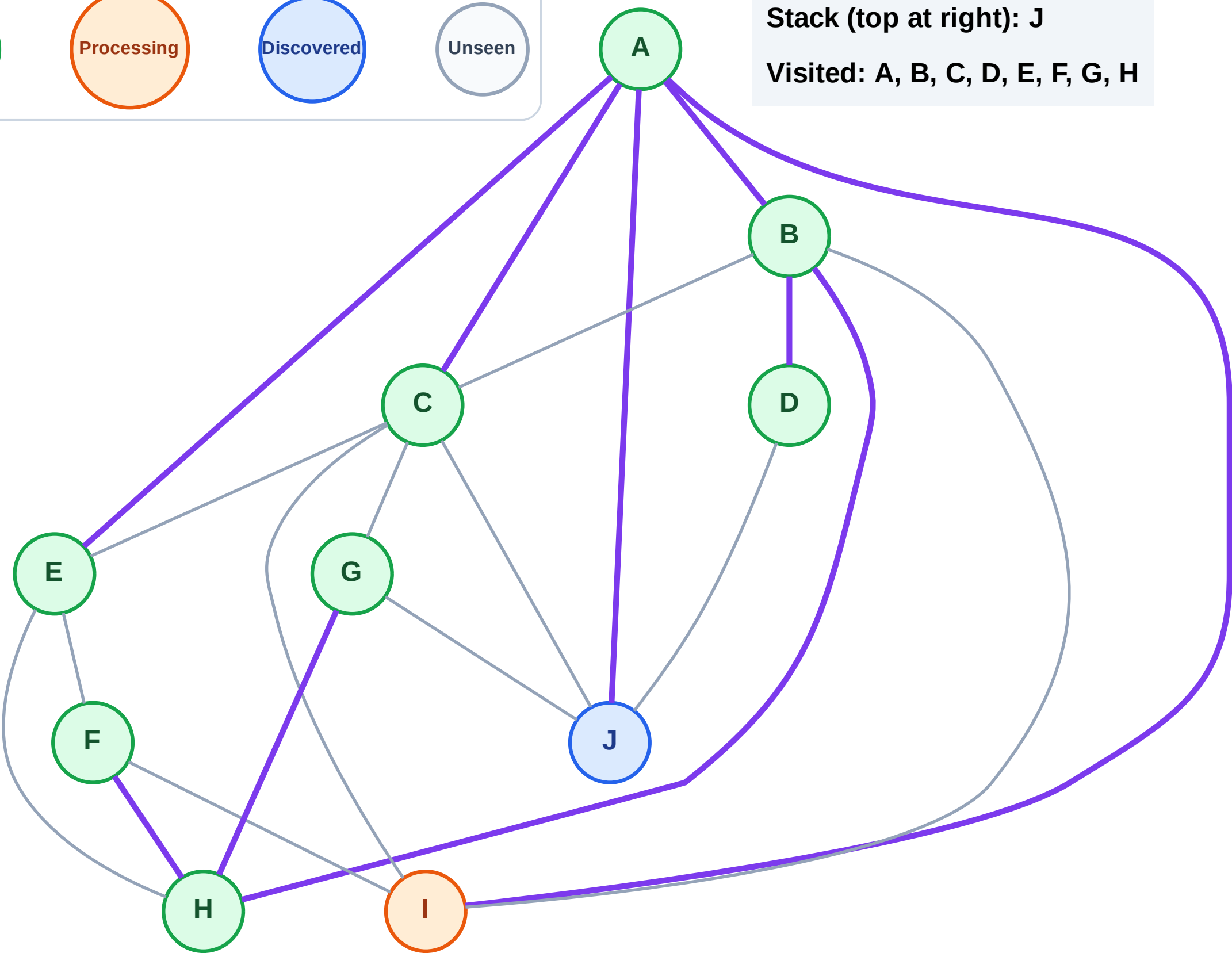
Step 18: Process node I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 19: No new neighbors from I

Legend

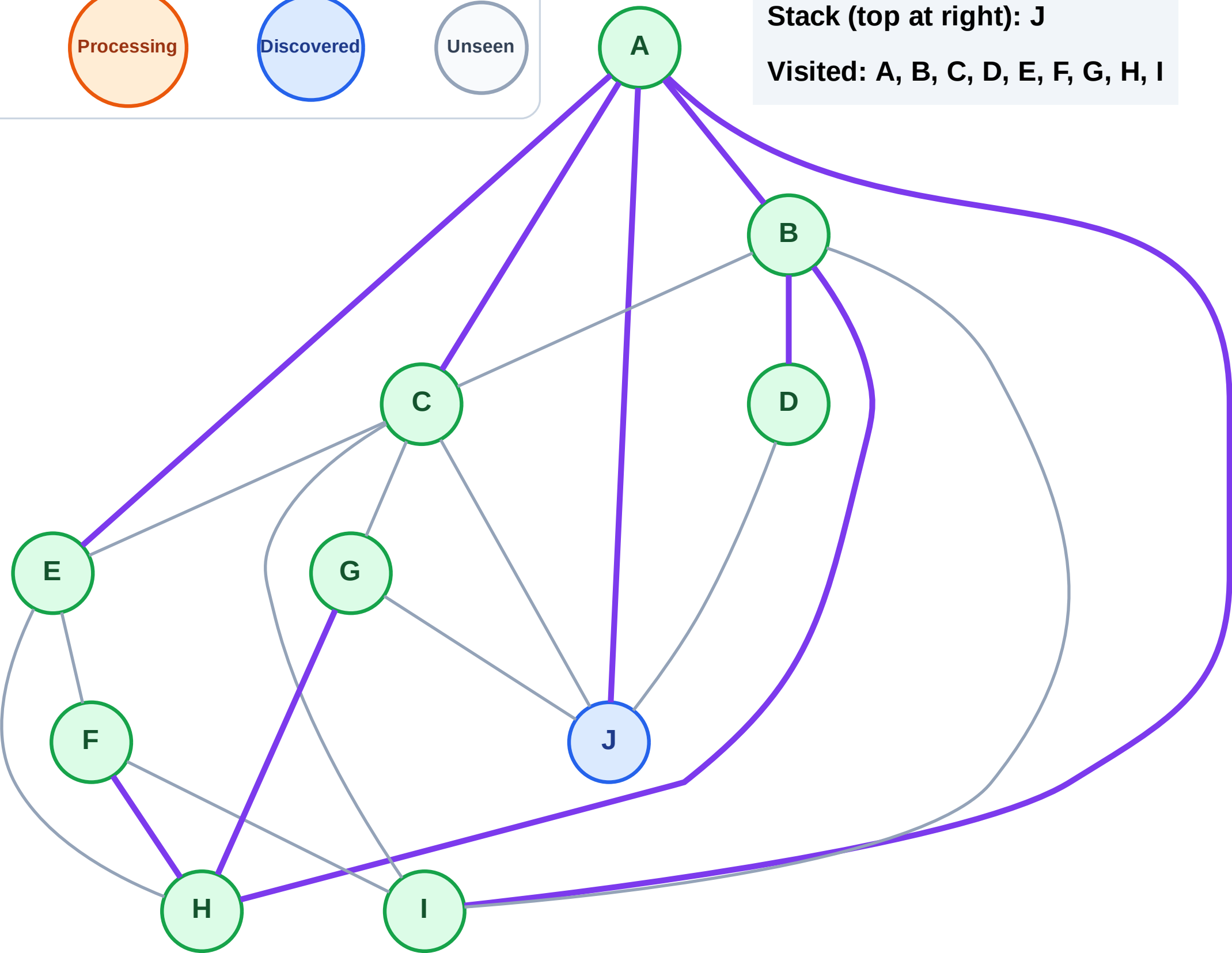
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

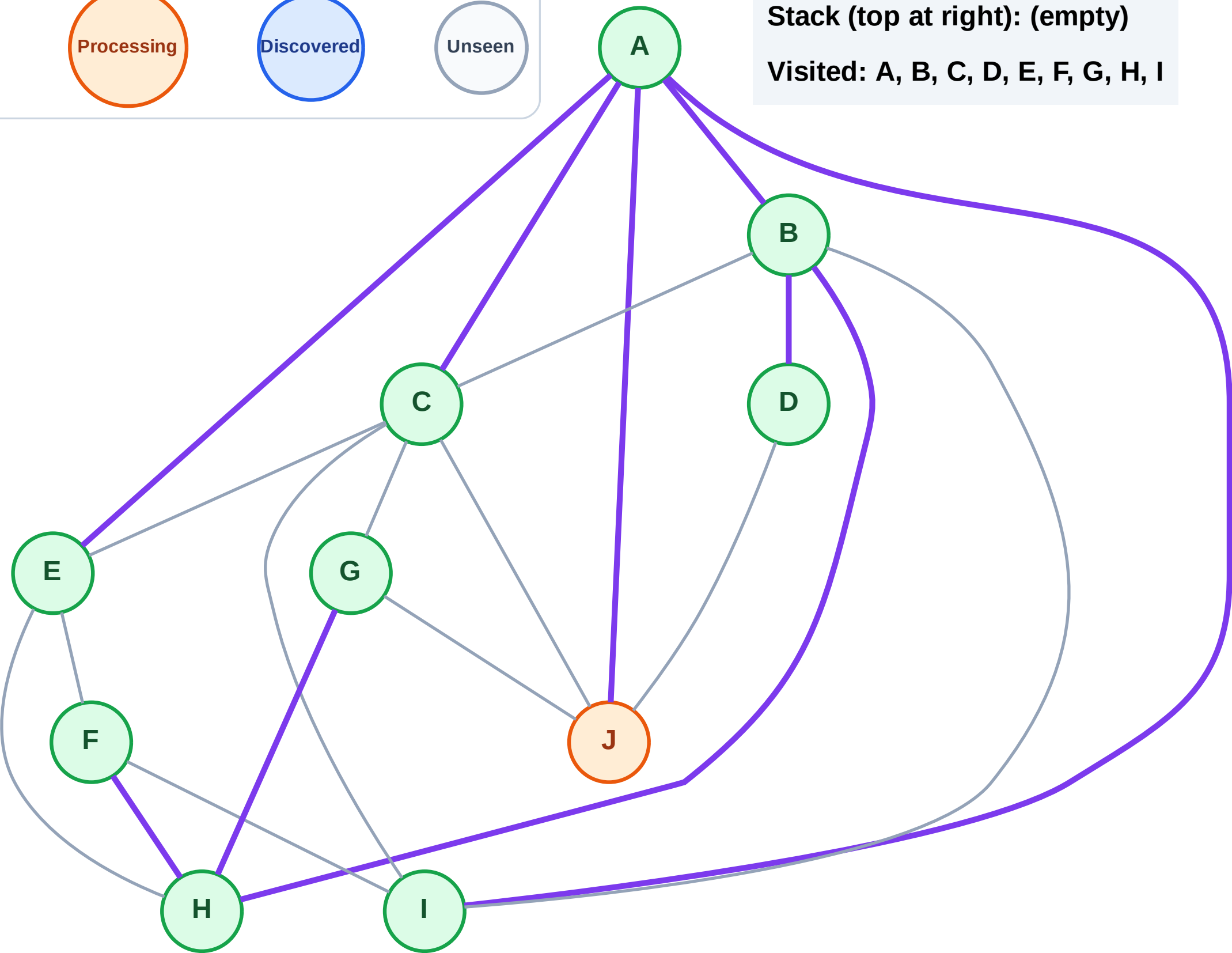
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

